CSCI 496: Senior Project Defense
Data Visualization for Election Administration
(Vote Data Center)

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Fall 2017

Statement of Purpose

Election administrators rely heavily on data. Voter registration, election results, precinct demographics, geographic data. All of these are vital to how we allot resources that will efficiently serve our voters. Unfortunately, this data is often hard to read and it is often impossible to understand trends or compute projections.

Data Visualization for Election Administration seeks to help Election Administrators make better use of the data we have available while at the same time show what further reports and data extraction might be necessary. The long-term goal of this project is to build user-friendly charts and graphs that is used to gather trends, identify areas of concern, and ultimately increase the accuracy of our resource management.

The current process of resource allocation and management relies simply on comparing spreadsheets, which include information such as turnout, equipment allocation, and other location-based data pertinent to the particular election. As you might imagine, this task is considerably time-consuming as well as often inaccurate. This is why the ability to not only visualize and describe the data, but also interpret the data is so important. This data visualization leads to more accurate analysis of election needs and provides a reliable base for constructing action plans.

I began with reports we currently use on a regular basis. These reports are generated in our statewide Voter Registration and Election Management System (VREMS), which includes data such as voter registration totals by precinct, registration locations, demographics, provisional ballot justification, absentee statistics, and other miscellaneous statistics. I also gathered data from various sources, such as our outreach department. I was then able to extrapolate the data into charts by utilizing JavaScript charting libraries in addition with PHP scripts, which could pull certain data from our internet sources and store them in a database. From there, I plot the data as well as allow the user to select which data points to view.

So far, the visualization of data has allowed our staff to make projections on upcoming elections more accurately and confidently. It also allows us to begin tracking registration statistics on desired intervals and create multi-year outlook projections. By utilizing multiple technologies, our Election Administrators are able to better serve the public as well as use this data in a meaningful way to influence the efficacy of voting and registration.

Research/Background

Our office (as well as other Election Administrators) have been in need of a dashboard-style interface, which allows the user to visualize various datasets easily in order to see trends and make projections on turnout or other events. There were also questions to answer, such as

1) Are there historical trends for precincts based on the election type?
2) What is a historical voter registration total trend?
3) Where should we have allocated more resources based on the election type?
4) At which locations is our outreach program most successful?

These, along with many other previously unknown questions, would be easily answered, visually, with the assistance of charts and graphs. Further, the ability to have these visualizations in a dashboard-style interface made it particularly convenient and efficient.

To accomplish this, the following was necessary:

1) Publically accessible dashboard interface.
2) Flexible data visualization options (ability to view various charts)
3) Ability to easily add both data and charts to dashboard from the back-end.
4) Charts that anyone can read and interpret.

Software Required for Project:

Numerous software systems were required for this project, including:

1) Access to the South Carolina Voter Registration and Election Management System (VREMS)
2) D3.js and C3.js JavaScript Data Visualization Libraries
3) JavaScript Map Plotting Library
4) Various scripting languages and tools such as PHP, HTML5, MySQL, JavaScript,

Data Visualization for Election Administration Test Plan

Introduction:

The testing phase for this project is multifaceted. First, we need to test the accuracy of the data entered – both manually and dynamically – into each chart. Second, the overall design code must be checked for compatibility and that it is free from errors. Third, we must test the security of the dynamic script, which connect to a MySQL database in order to pull data.

References:

- We must test each report that is used to generate data against the final chart output. These reports with the exception of the last six URL's are behind a secured web portal only available to South Carolina Election Administrators. The used charts for this project are:
    - (RP0028) Audit Trail Report
    - (RP0040) Registration Stats by Race/Precinct/Gender/Age
    - (RP0041) Registration Stats by District/Precinct/Race
    - (RP0042) Registration Stats by Race/Precinct/District
    - (RP0043) Registration Stats by Precinct/Race
    - (RP0107) UOCAVA Statistics by Military/Citizen for Election
    - (RP0108) UOCAVA Statistics by Military/Citizen for Primary
    - (RP0123) Voter Registration Totals by County, Precinct and District Type
    - (RP0126) Voters Moved Between Counties
    - (RP0128) Online Registrations by Type
    - (RP0130) UOCAVA Voters by Ballot Style
    - (RP0021) Participation Stats by Registration Location
    - (RP0114) In-Person Absentee Voting Log
    - (RP0121) State Reimbursement Summary Cover Letter
    - (RP0134) Provisional Ballot Statistics
    - (RP0052) Motor Voter Registration Location Statistics
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen06&regvot=VOT
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen08&regvot=VOT
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen10&regvot=VOT
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen12&regvot=VOT
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen14&regvot=VOT
    - https://www.scvotes.org/cgi-bin/scsec/vothistcty?ctynam=CHARLESTON&election=vhgen16&regvot=VOT

Features to be Tested:

- Accuracy of entered data.
- Compatibility of PHP and CSS code with various browsers including Firefox, Chrome, Safari, and Internet Explorer.
- Data selection inputs on charts that allow switching out data.
- Overall Layout for entire site.

Approach:

- The overall approach to testing is simple. First, we create required report from VREMS, and then visually compare the report data with the chart output. For larger datasets, random "spot checking" will suffice – preferably samples from the beginning, middle, and end of the dataset.
- To verify code compatibility, an online tool (http://webpagetest.org) is used.
- For data selection, a similar approach to the overall testing is used. However, new reports must be generated to match selected dataset.
- The overall site layout is a visual inspection to ensure charts are properly aligned, margins are correct, all options (such as expanding charts) work properly, and that there are no errors in the browser Console.

Item Pass/Fail Criteria:

- Due to the "informational purposes only" nature of the website, 100% accuracy is not required; however, data must line up to appropriate categories.
- In addition, all PHP, HTML, and JavaScript must work flawlessly and throw no errors.
- For security, applicable code must pass

Test Environment:

- Test environment 1:
  - Windows 7, 64bit
  - Chrome, version 62.0.3202.94
  - Internet Explorer. Version 11.0.9600.18762
  - Firefox, version 57.0
- Test environment 2:
  - macOS 10.13 High Sierra
  - Safari

Assumptions:

- Pertaining to the chart data - the major assumption for the testing we have is that the sources data provided in the reports is accurate.

Challenges Overcome

During this project, there were many challenges to overcome. Originally, I had hoped to be able to provide automatic daily registration data. Unfortunately, the reports are behind a secure portal (VREMS) and I cannot gain access with scripts. In addition, SCVotes.org only updates registration totals quarterly. To overcome this, I have to manually create a report inside VREMS and copy the information into the JavaScript file.

Due to advancements in technology in elections, namely the creation of VREMS, over the past 5 years, there is limited electronic data prior to 2012. However, of the reports that are available, there are limited detailed reports showing everything one would ask.

Future Enhancements

Going forward, Vote Data Center will expand into more charts and graphs as necessary. I would also like to work alongside the State Election Commission to gain access to VREMS via API or some other method. This will allow me to gather live or dynamic data including up-to-the-minute absentee statistics and voter registration totals.

This project also allows for the expansion into statewide data, not just Charleston County. The ability of allowing the public to request data and charts will also be an option.

Main Dashboard

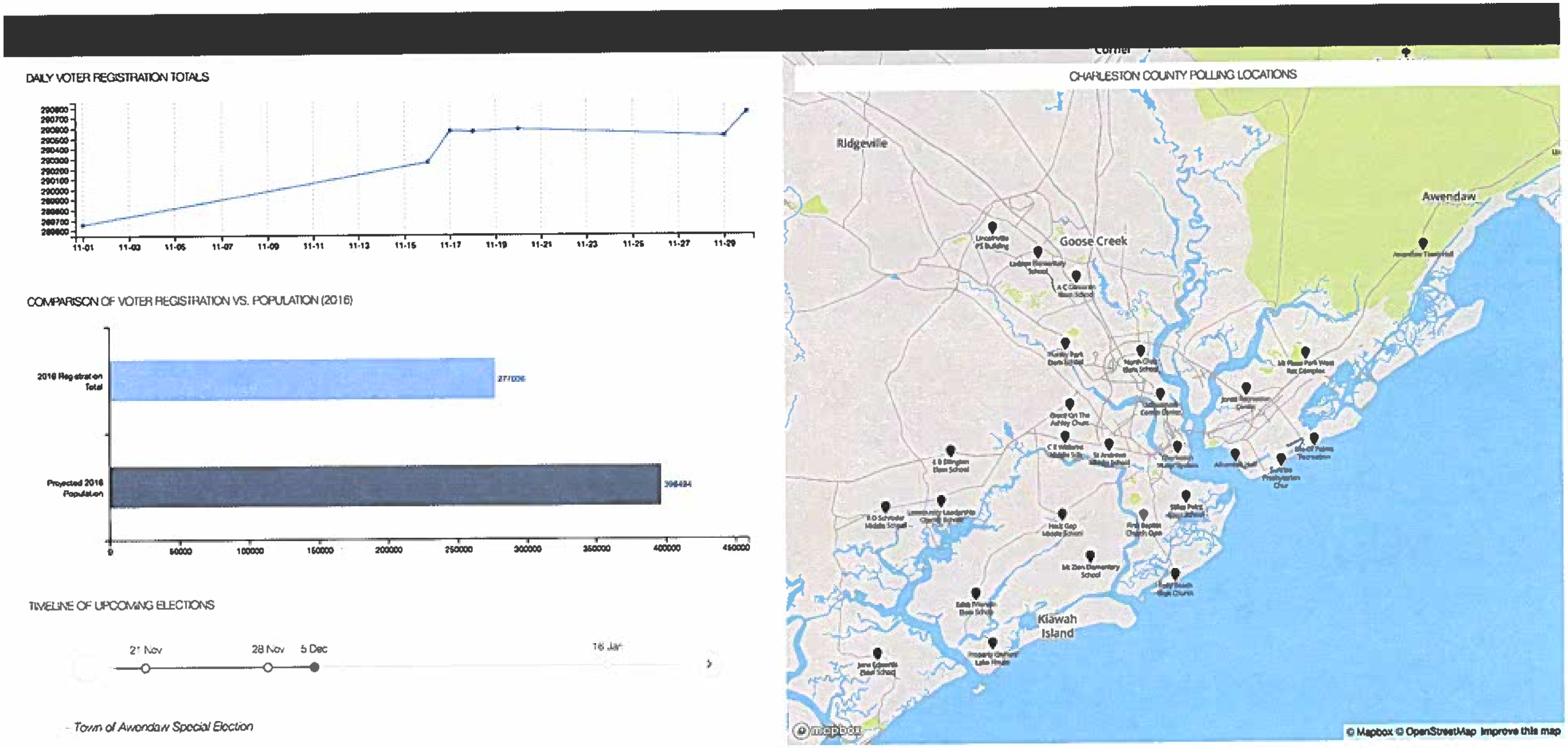


Figure 1: Main Dashboard

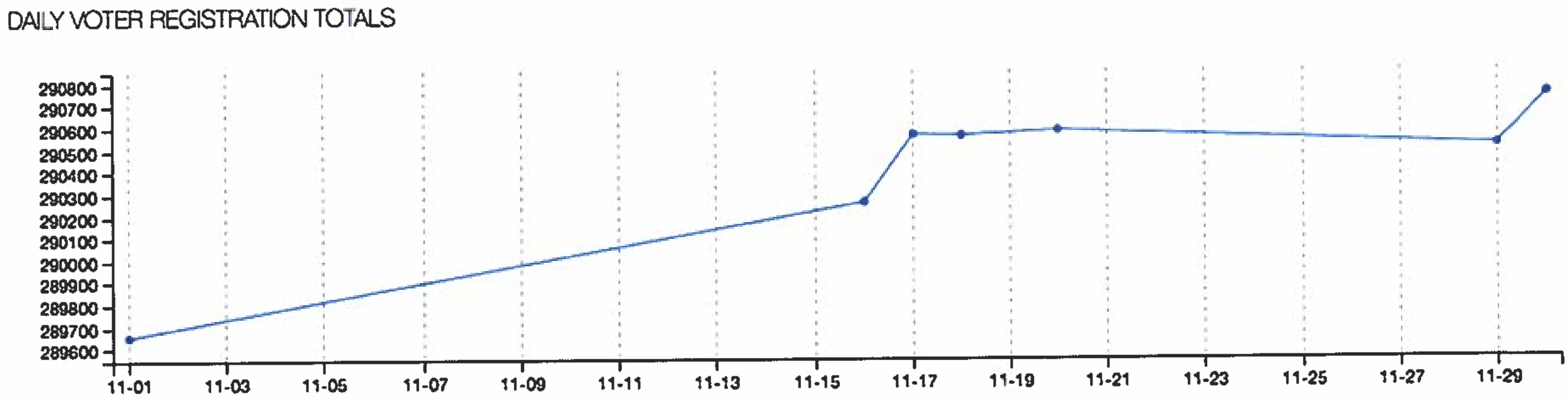


Figure 2: Registration Total on Intervals

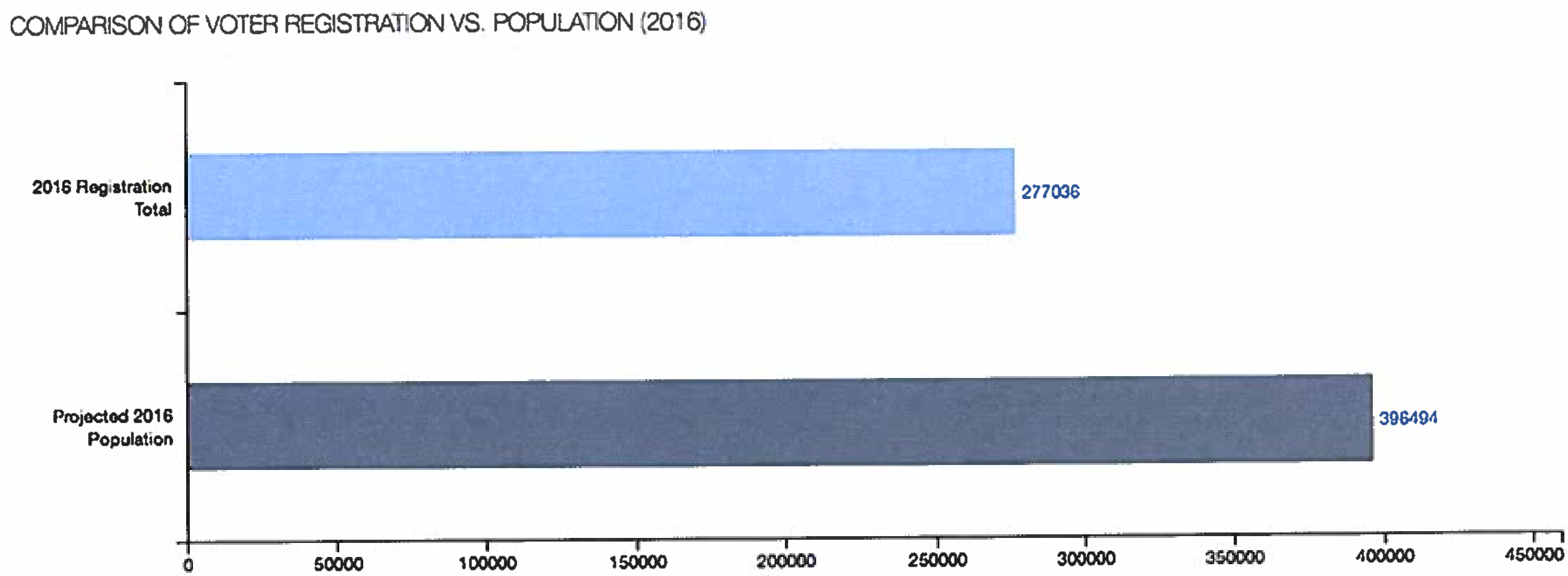


Figure 3: Registration vs. Population Comparison

Figure 4: Timeline of Elections

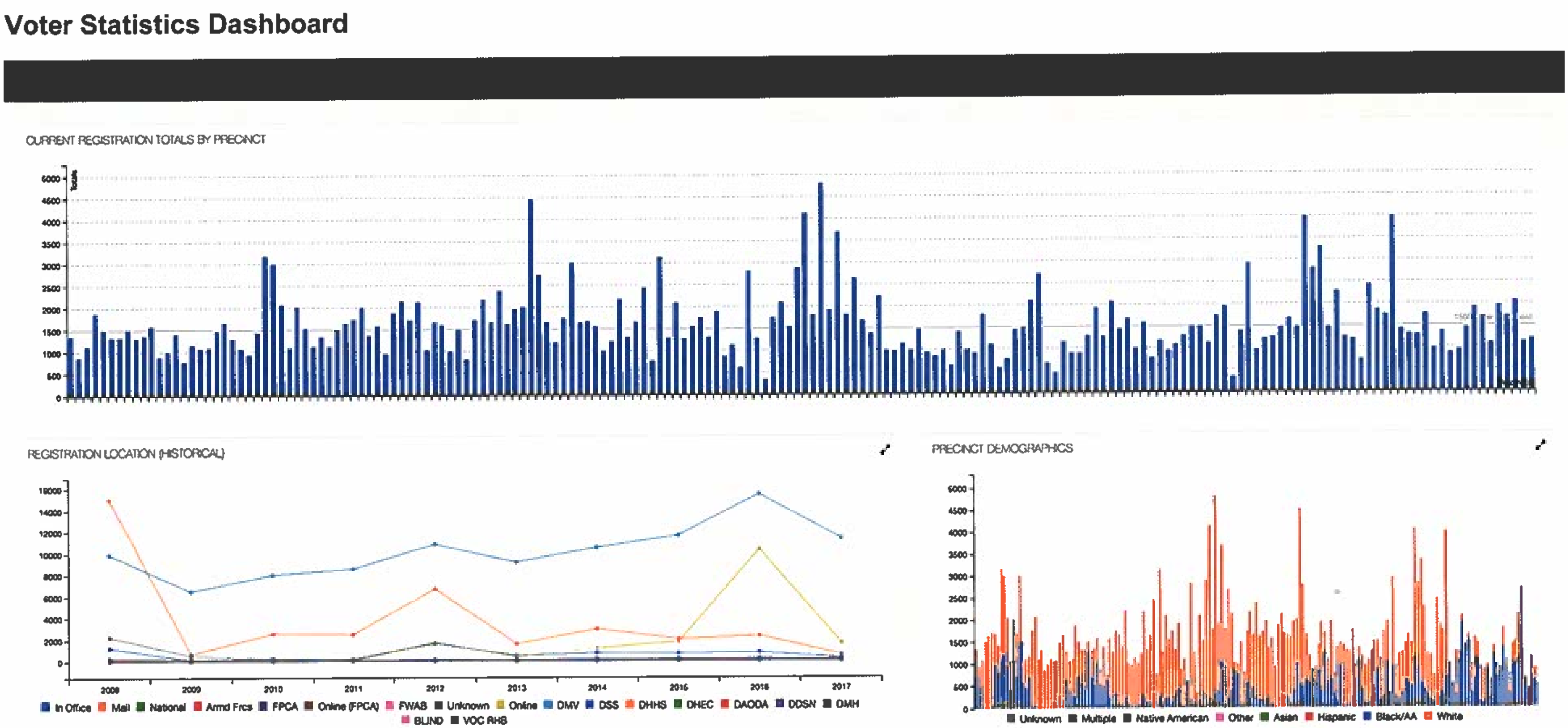


Figure 5: Voter Statistics Dashboard

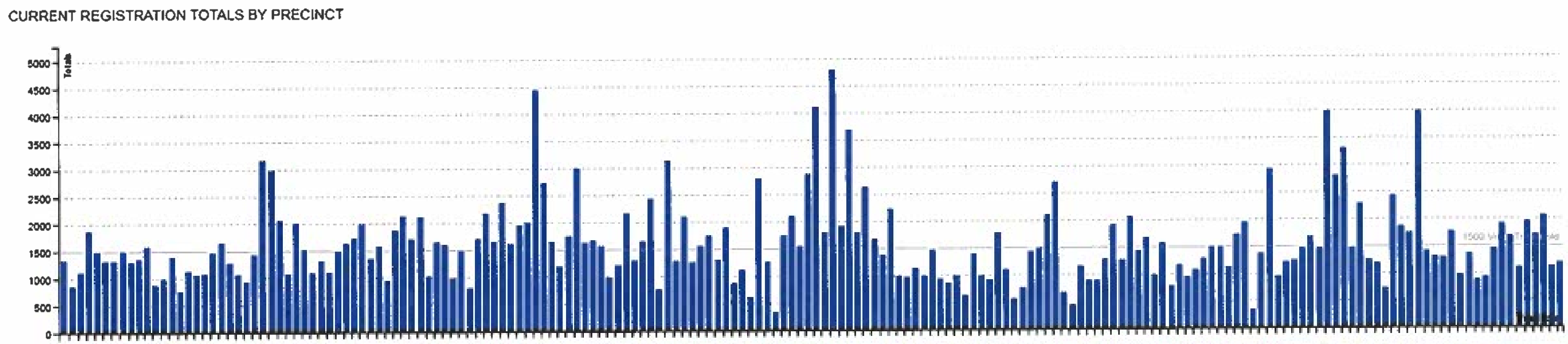


Figure 6: Registration by Precinct (with 1500 limit line)

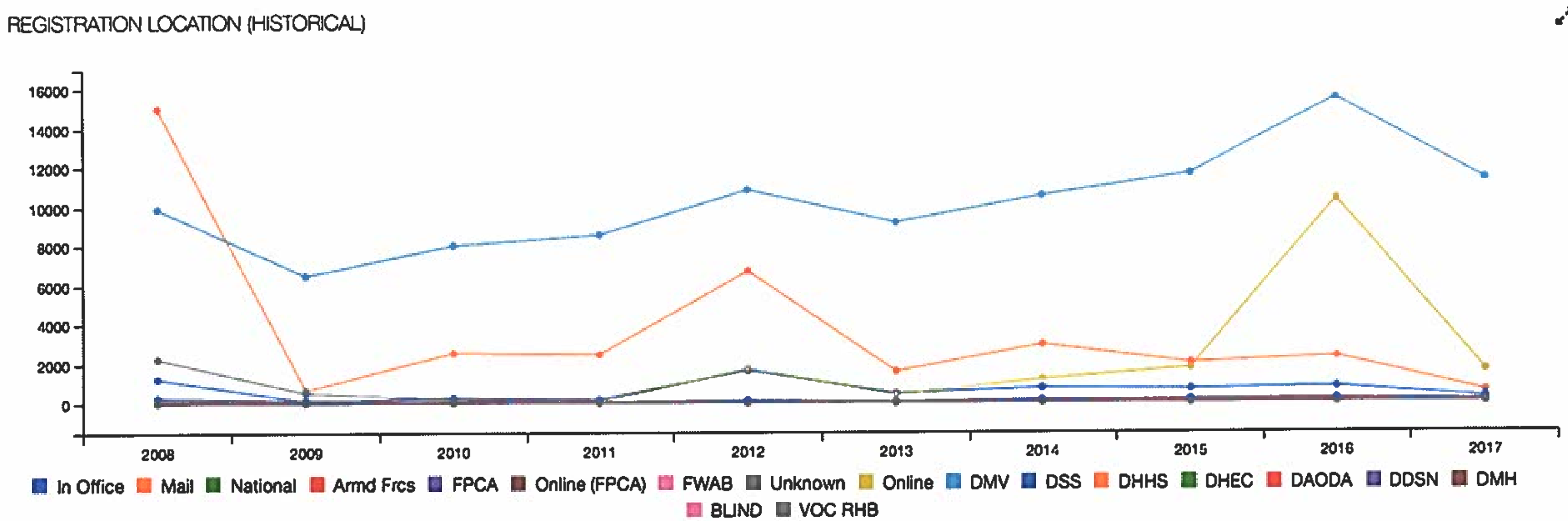


Figure 7: Historical Registrations by Location

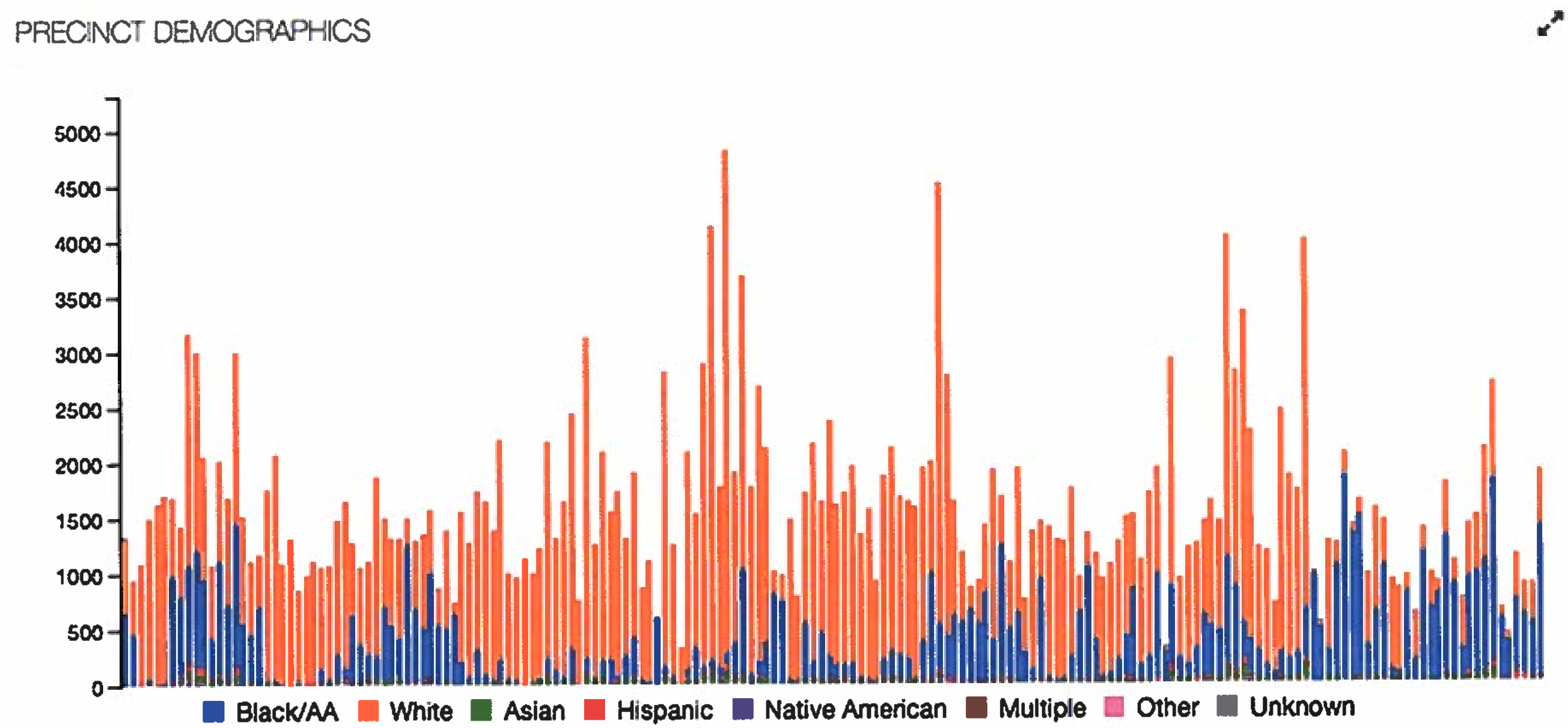


Figure 8: Current Precinct Demographics

Voter Mapping Interface

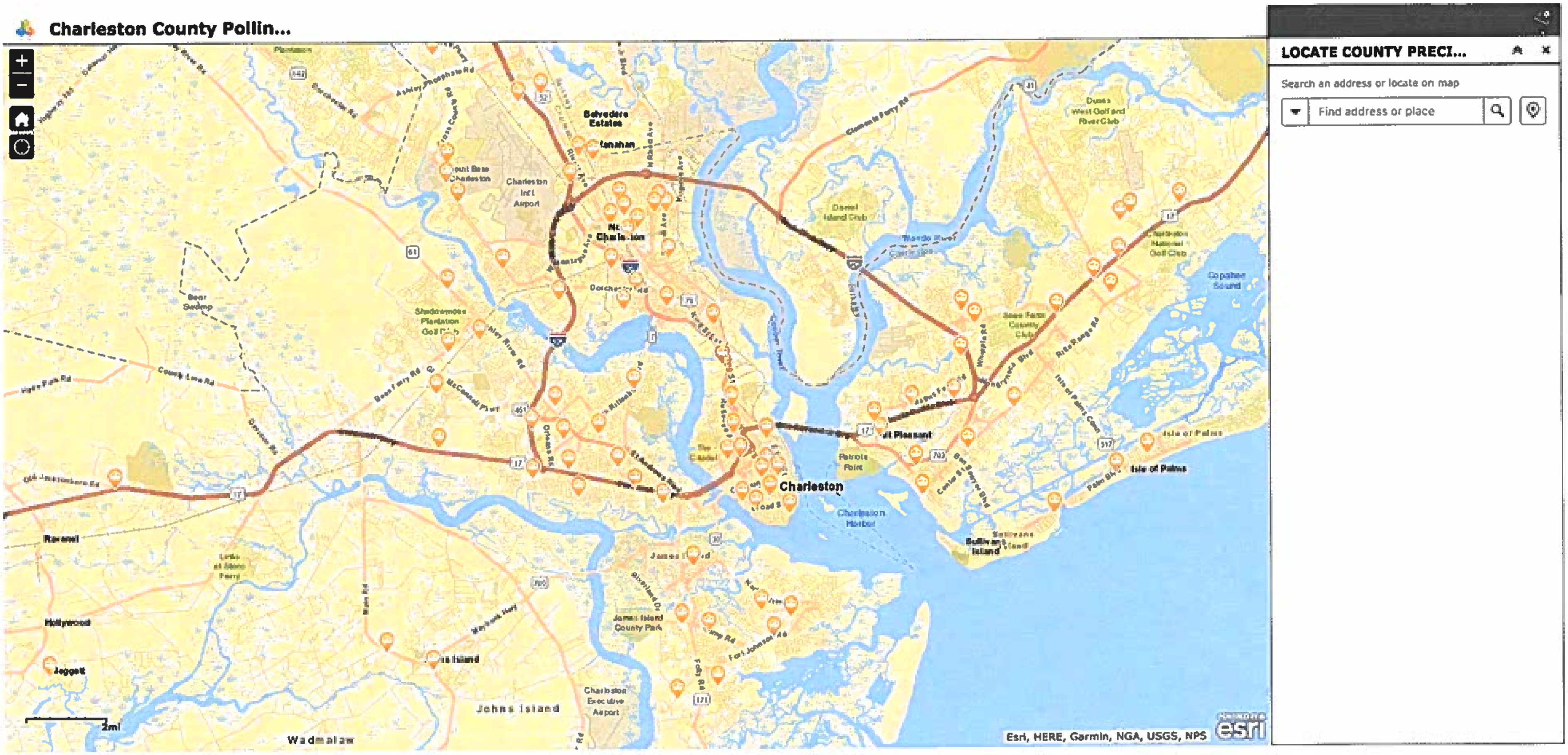


Figure 9: Voter Mapping Interface

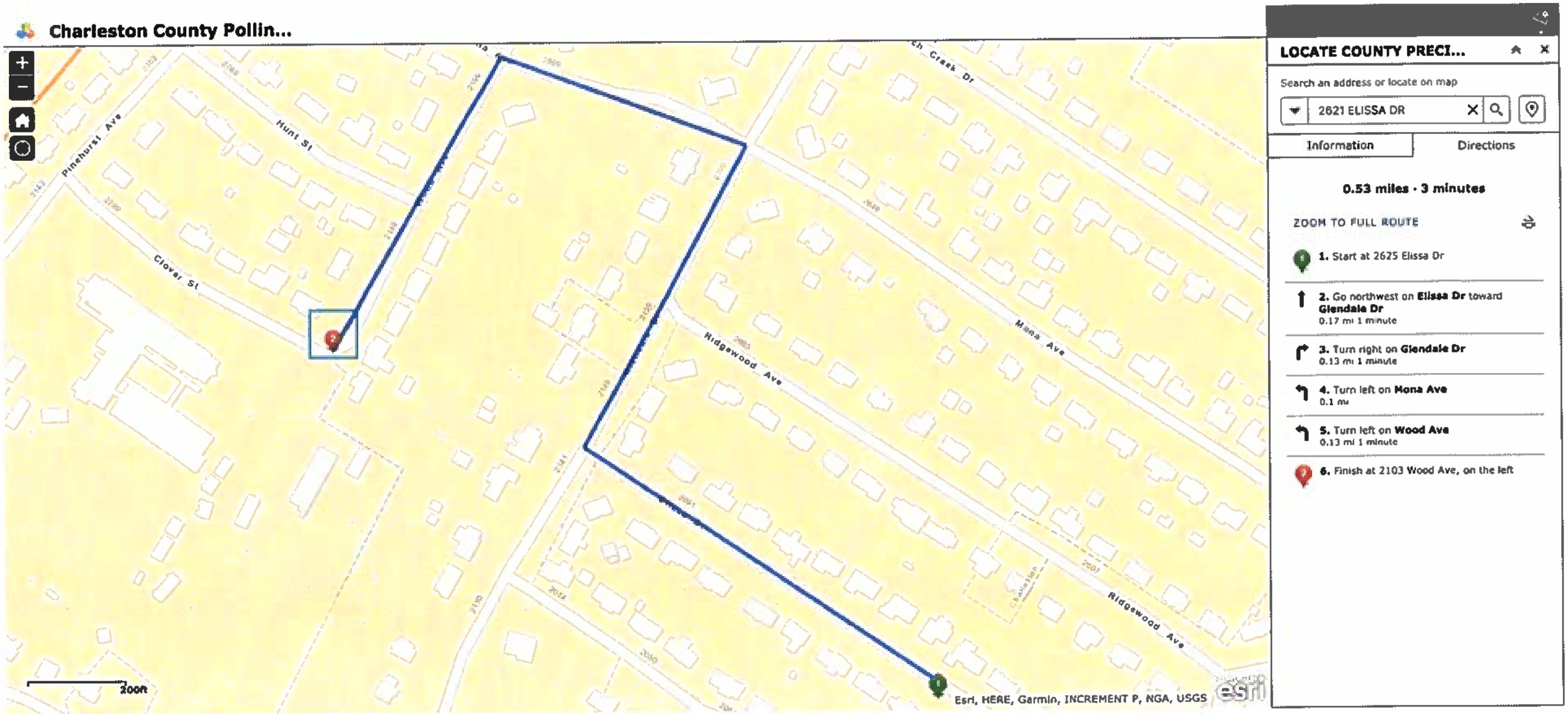


Figure 10: Directions via Voter Mapping Interface (Charleston County GIS)

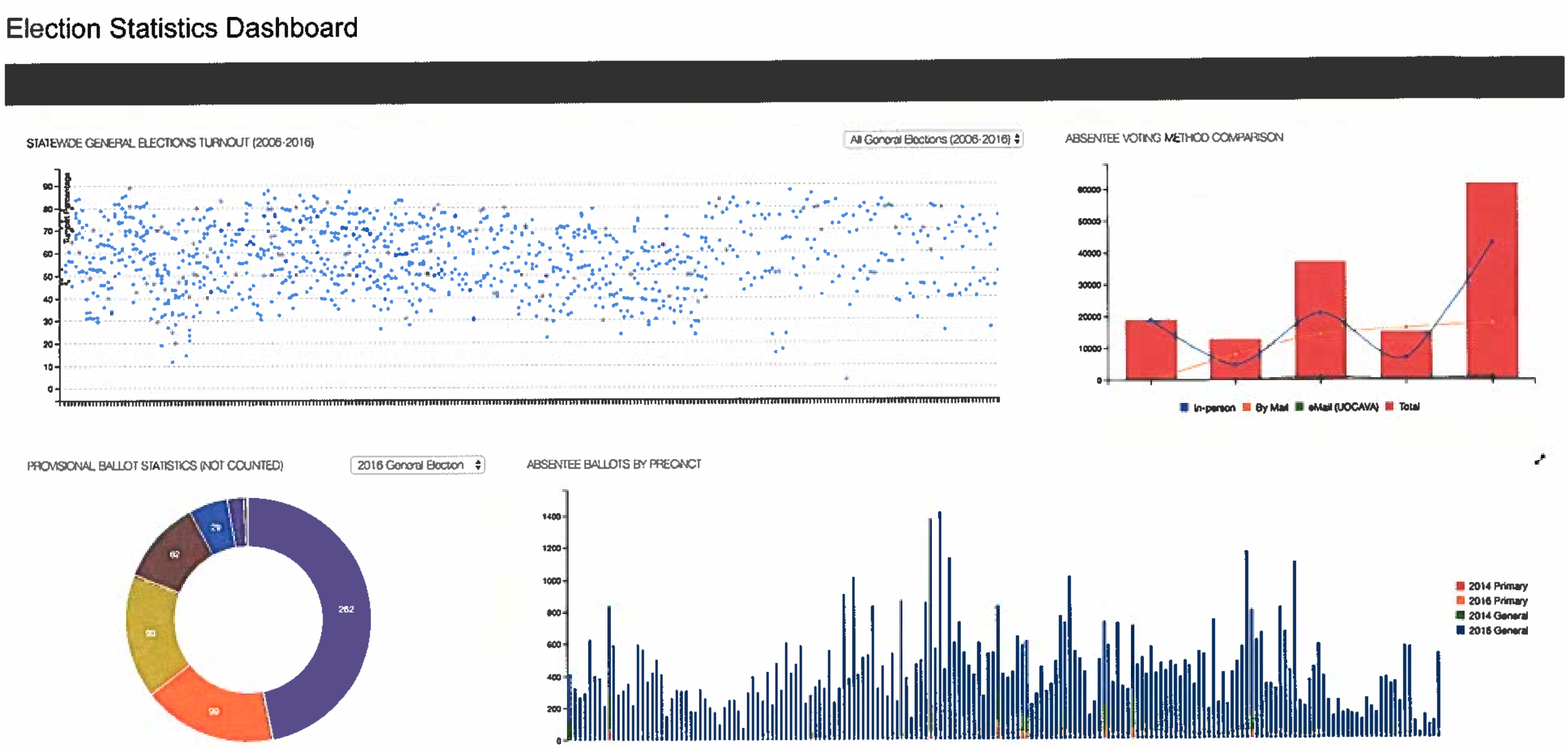


Figure 11: Election Statistics Dashboard

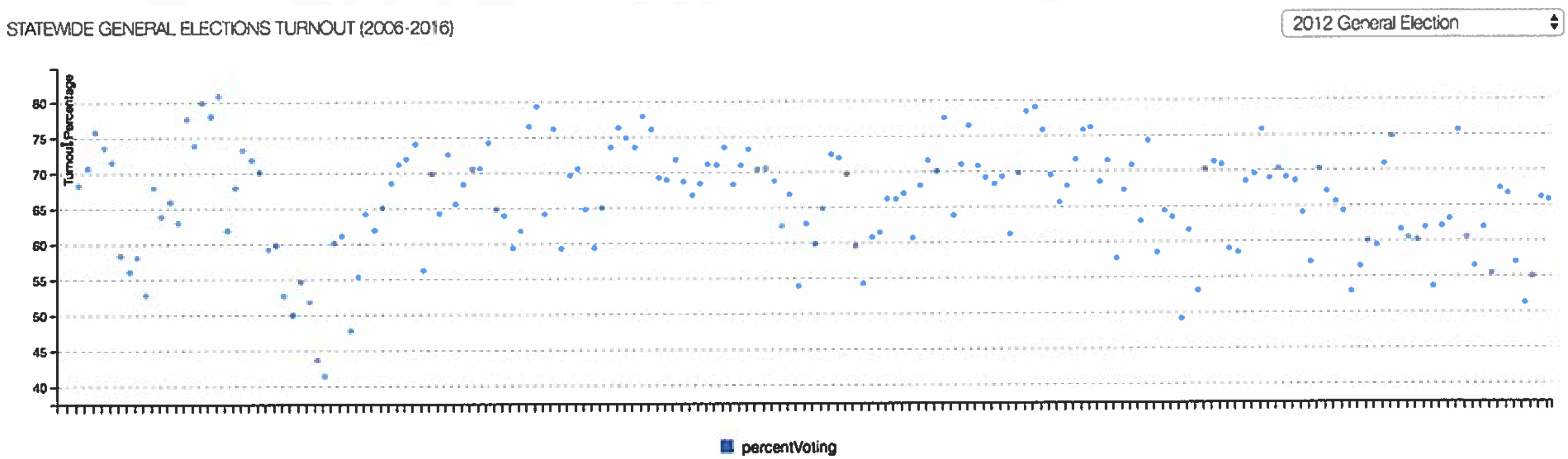


Figure 12: Turnout Scatterplot (showing selected election)

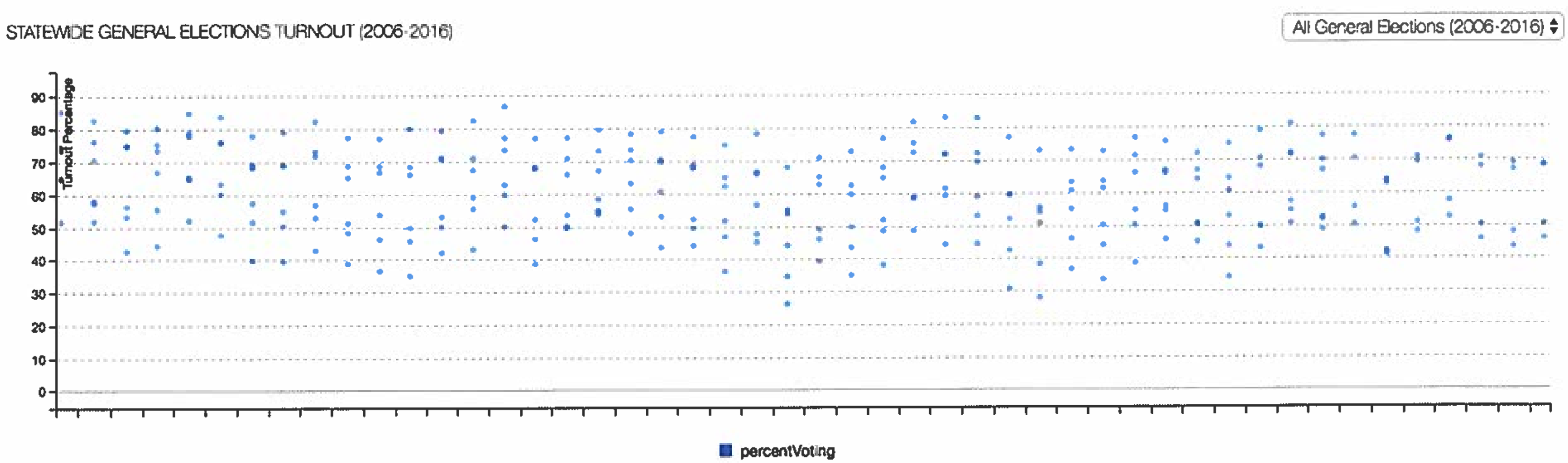


Figure 13: Turnout Scatterplot (zoomed)

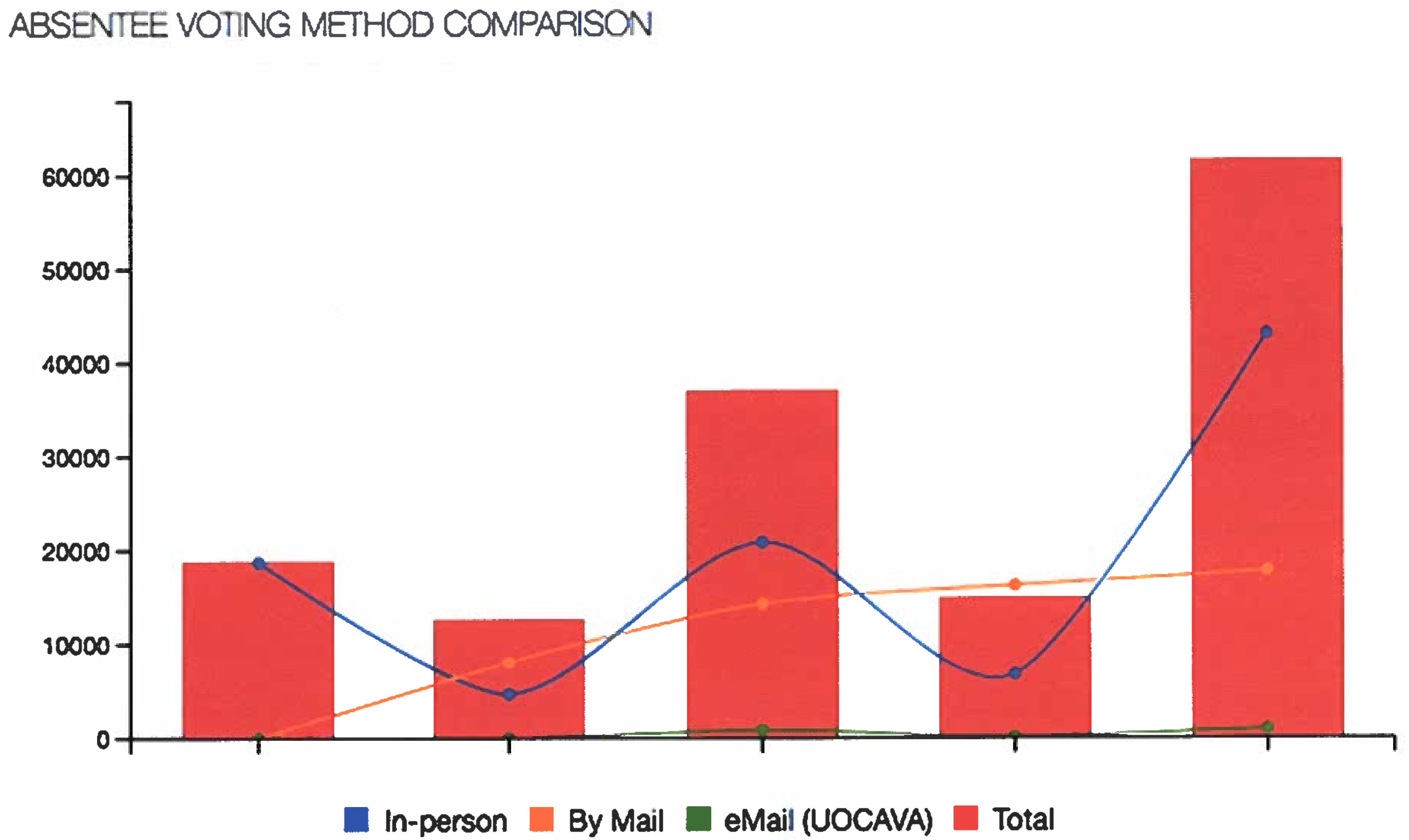


Figure 14: Absentee Voting Method Comparison (2008-2016 General Elections)

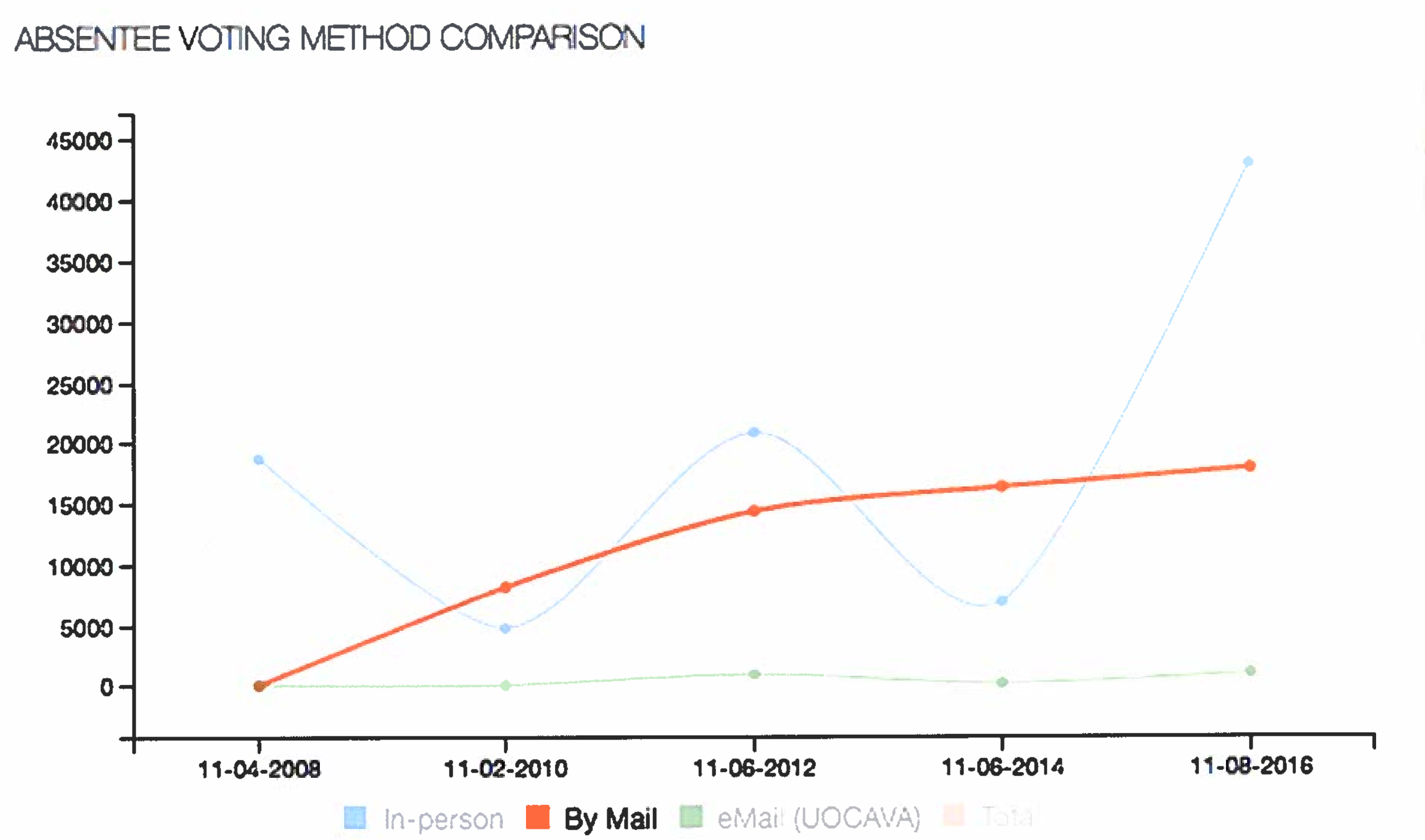


Figure 15: Absentee Voting Method Comparison (Selected option for trend)

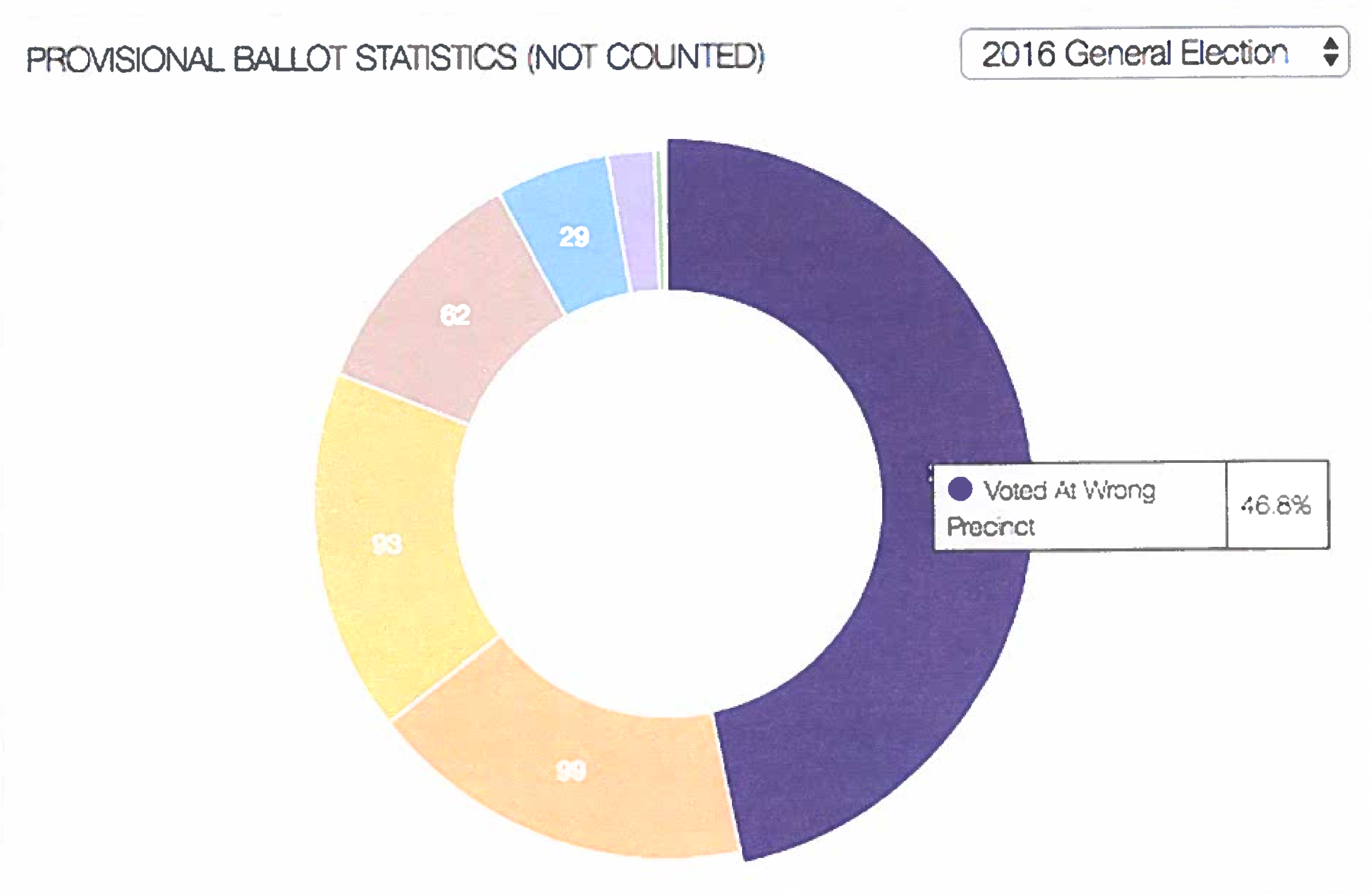


Figure 16: Provisional Ballot Statistics (Not Counted - with reasons)

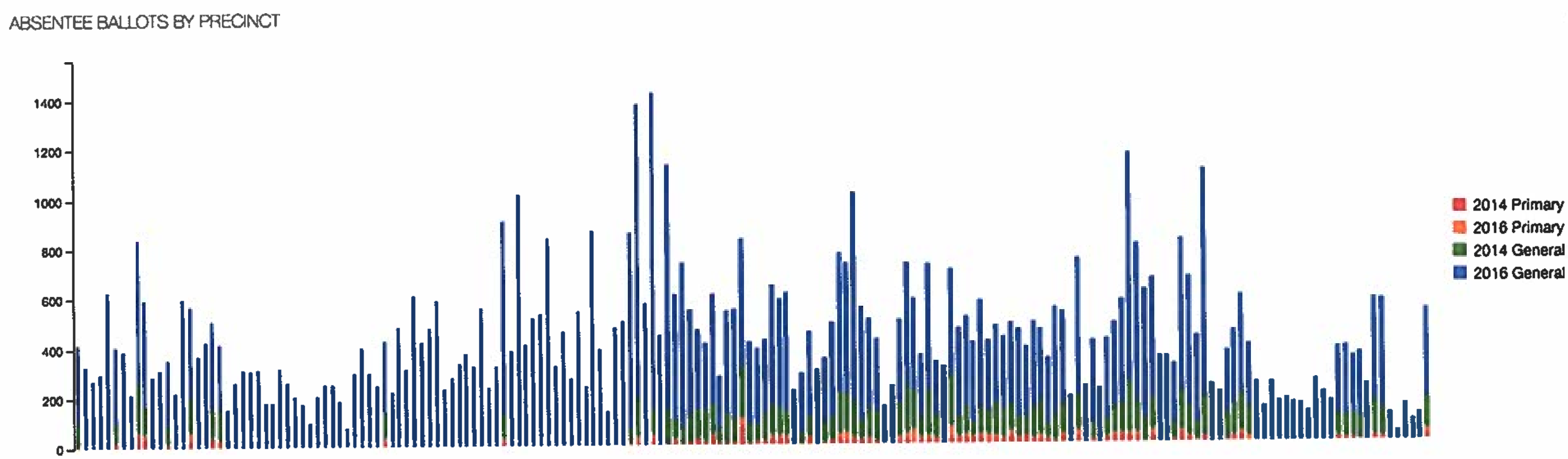
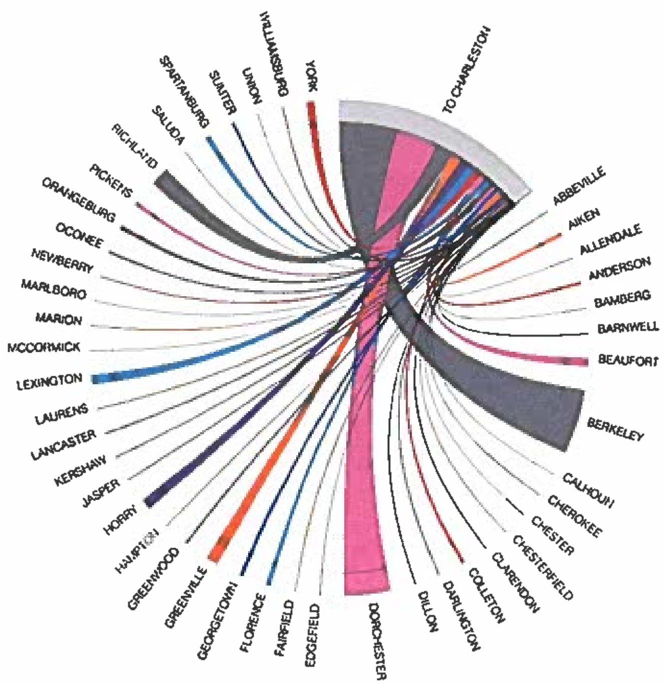


Figure 17: Absentee Ballots by Precinct

Data Mining Dashboard

VOTERS MOVED TO CHARLESTON (2017)



PARTICIPATION BY REGISTRATION LOCATION

2016 General Election

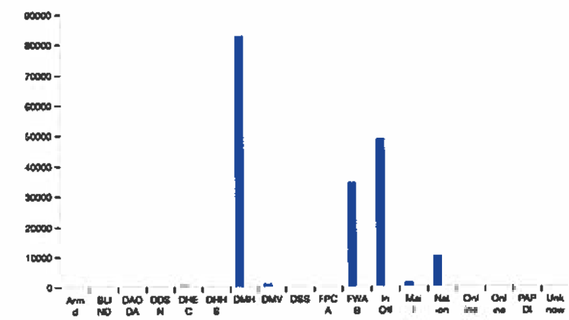


Figure 18: Data Mining Dashboard

VOTERS MOVED TO/FROM CHARLESTON (2017)

Voters Moved Into Charleston

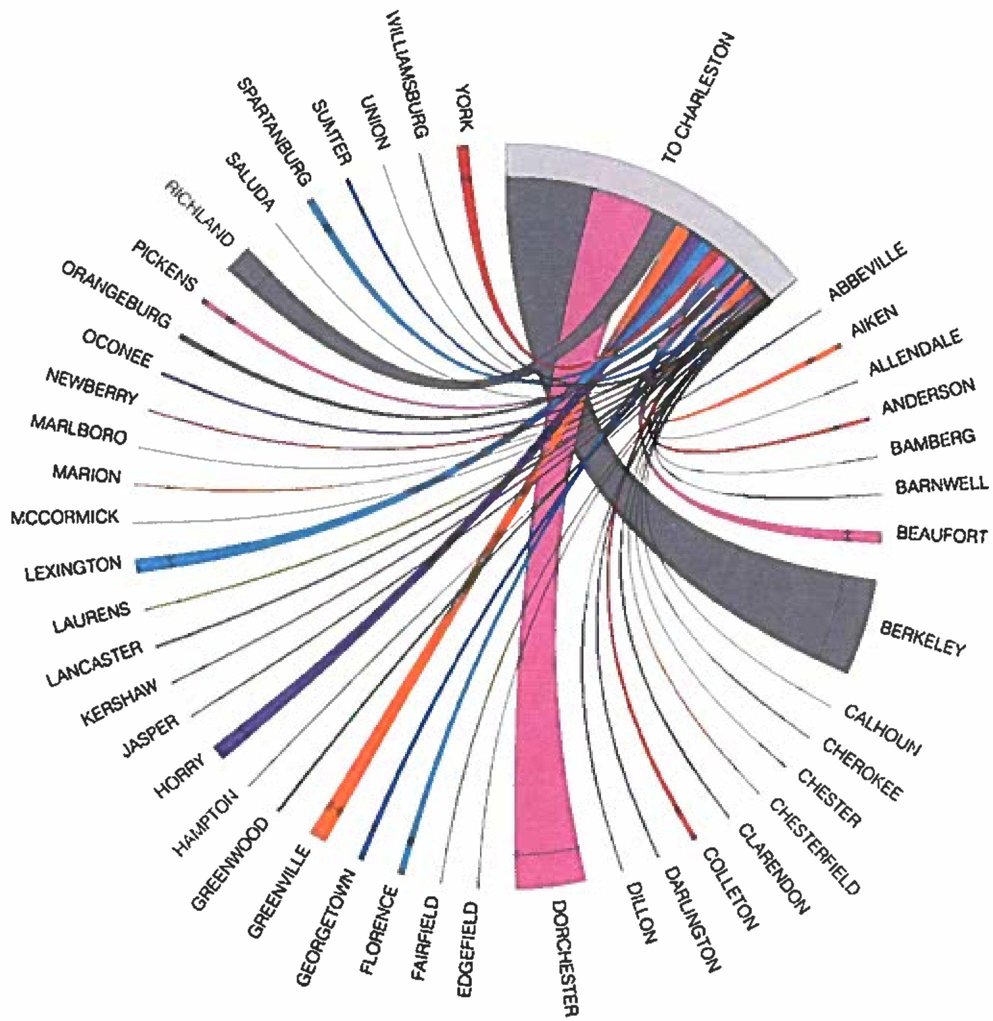


Figure 19: Chord Chart showing voters who moved to Charleston (Relationally with other counties)

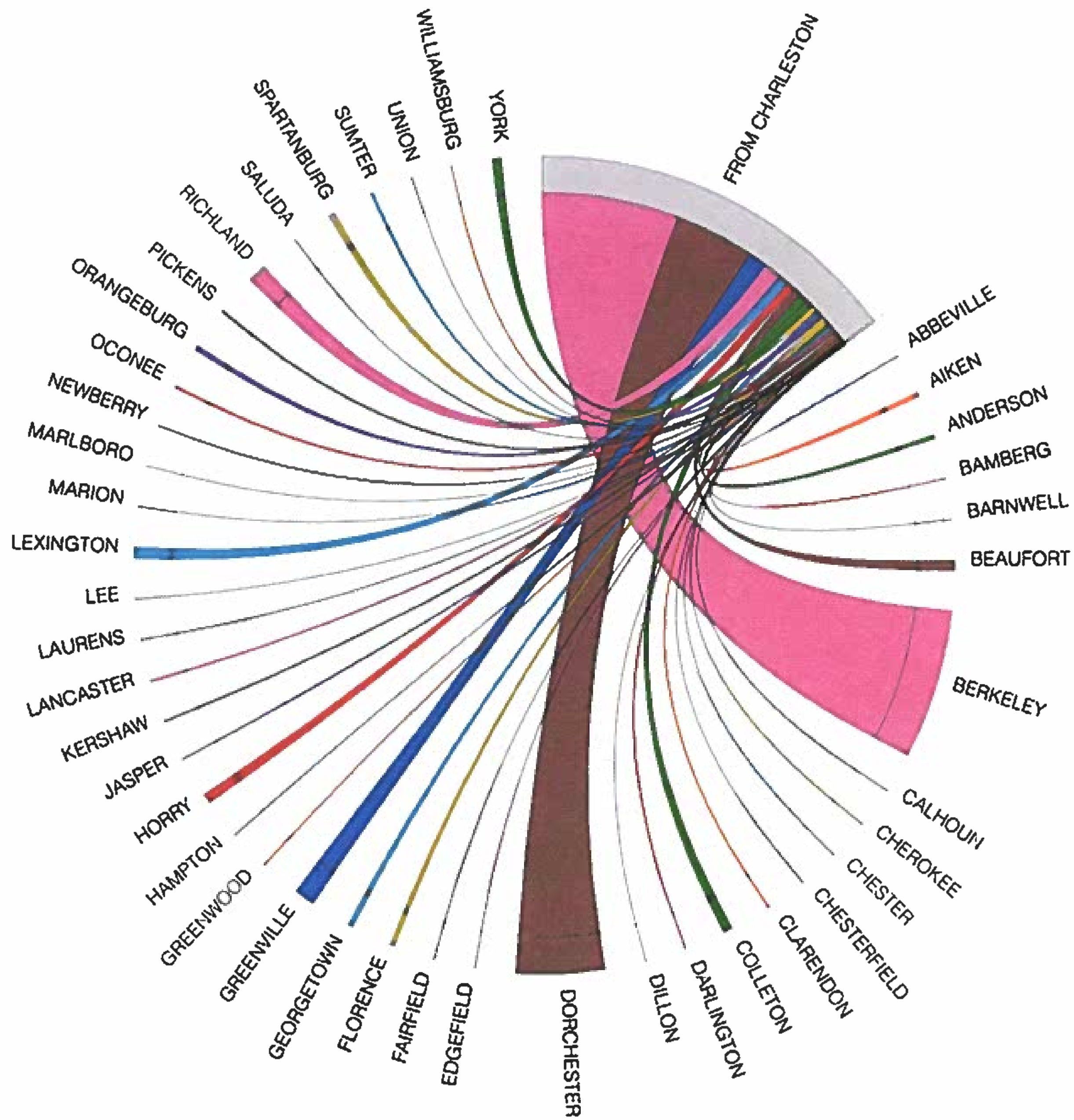


Figure 20: Chord Chart showing voters who moved from Charleston

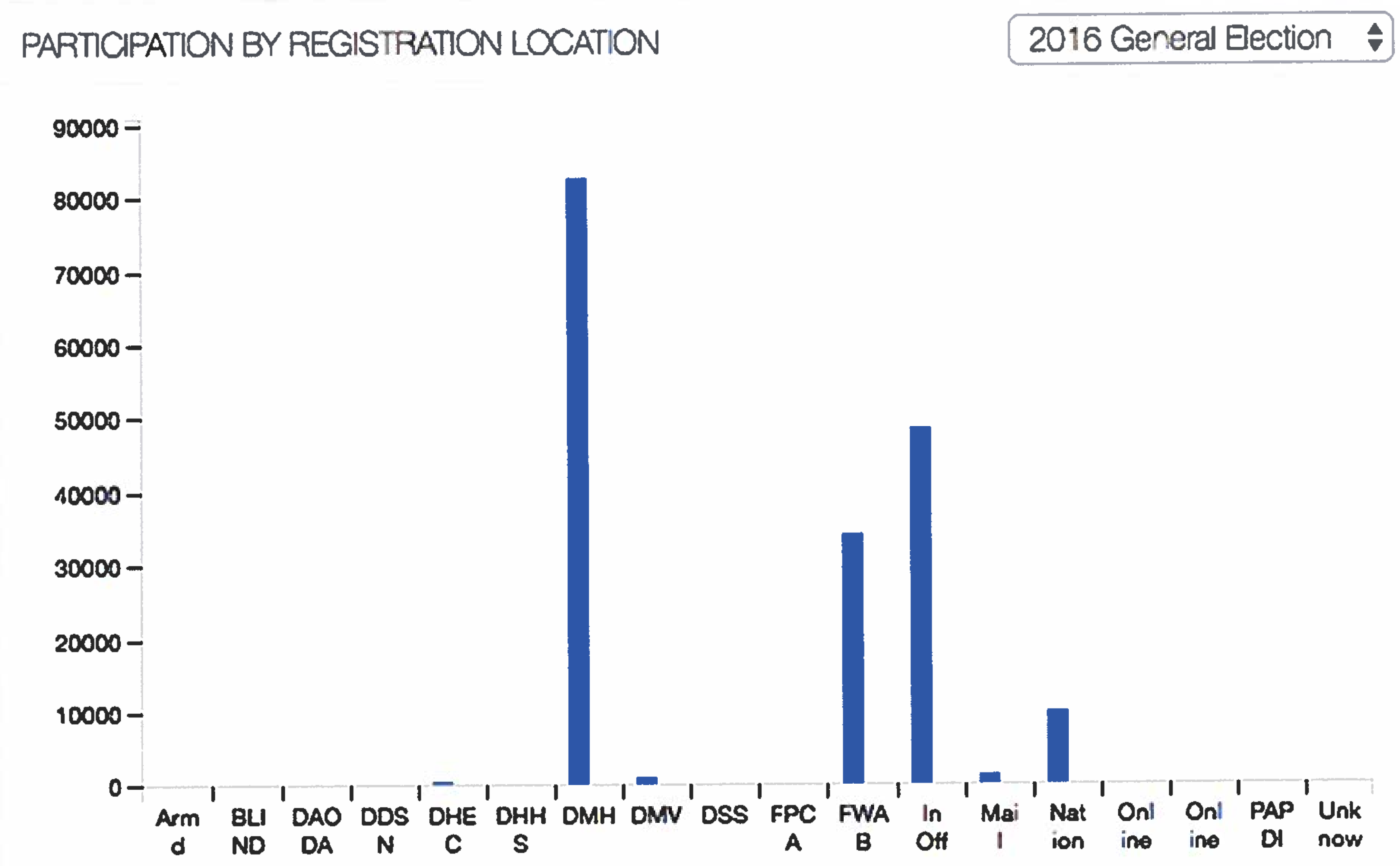


Figure 21: Bar chart showing Participation by Registration Location

Expandability

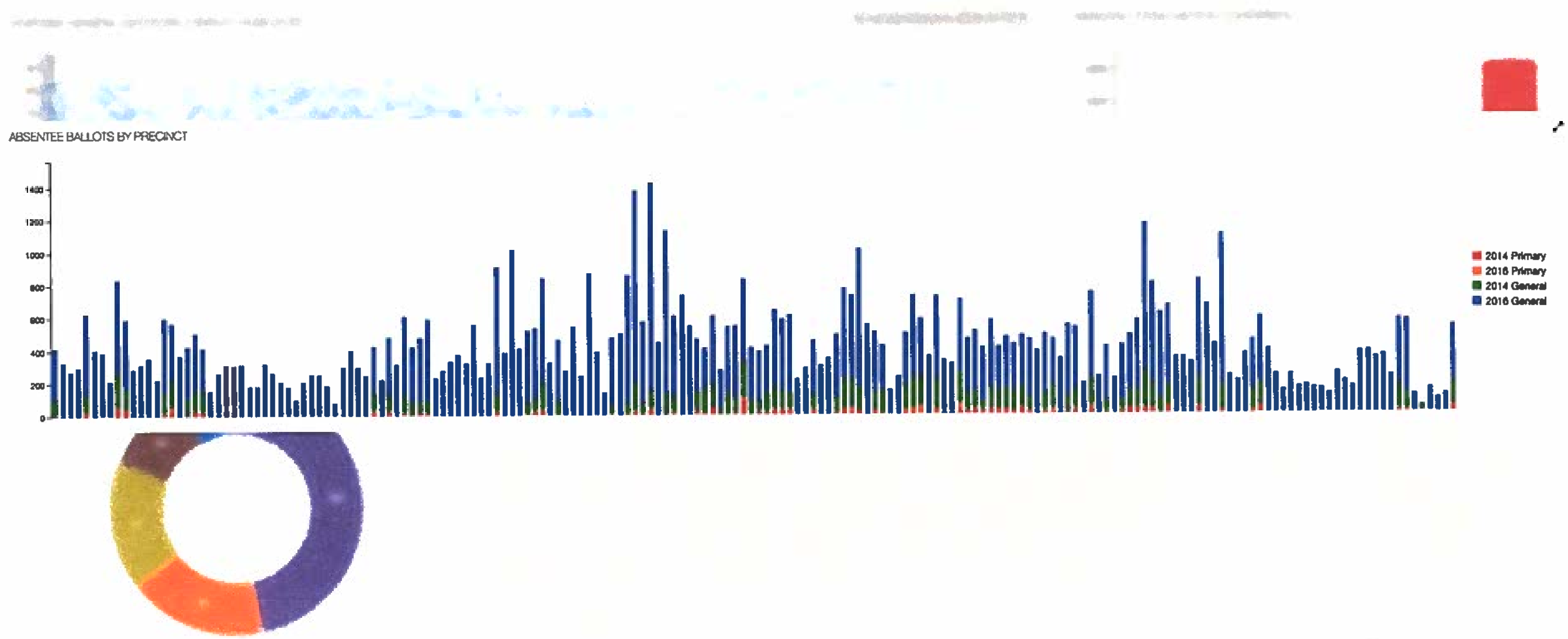


Figure 22: Expandable Charts

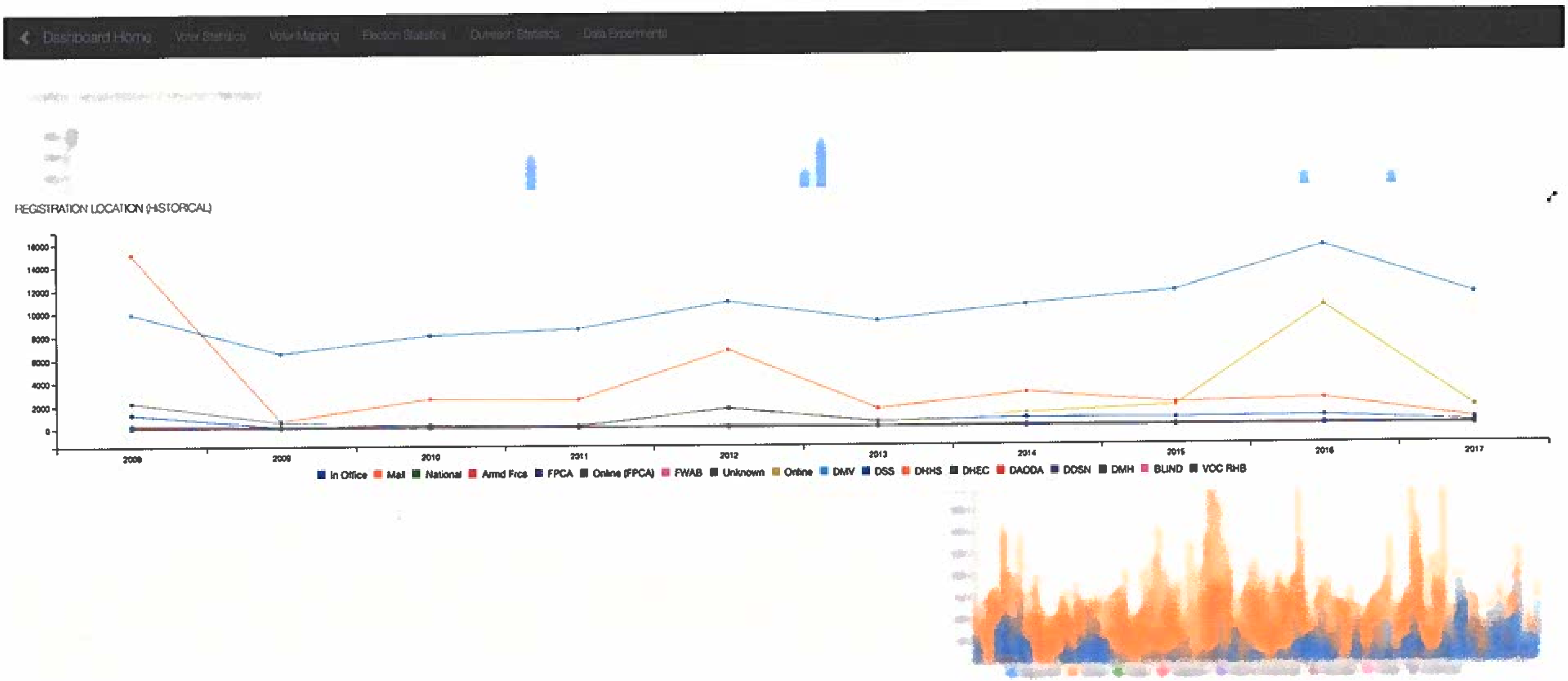


Figure 23: Expandable Charts

Data Visualization for Election Administration

(VoteDataCenter.us)

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Fall 2017

The Problem

- Voter/Election data is currently available to view in spreadsheet, flat text, or some other text-based format.
- Hard to read.
- Almost always impossible to see relationships.
- <u>Simple charts for simple people</u>
  - Dashboard-style interface
  - Publically accessible

Challenges

- Limited availability of detailed reports.
- Lack of electronic data prior to 2012.
- No ”live” data.
- Terribly formatted reports.
- Manual entry required for most data.

The Solution

- South Carolina Voter Registration & Election Management System (VREMS)
- C3.js & D3.js Javascript Data Visualization Libraries
  - Provides the back-end for data-driven documents.
- Mapbox location data platform
- Other web technologies:
  - jQuery, HTML5, CSS3, mySQL, PHP, etc.

Data-Driven Documents

C3.js D3-based reusable chart library

mapbox

jQuery write less, do more.

HTML

CSS

The Old Way

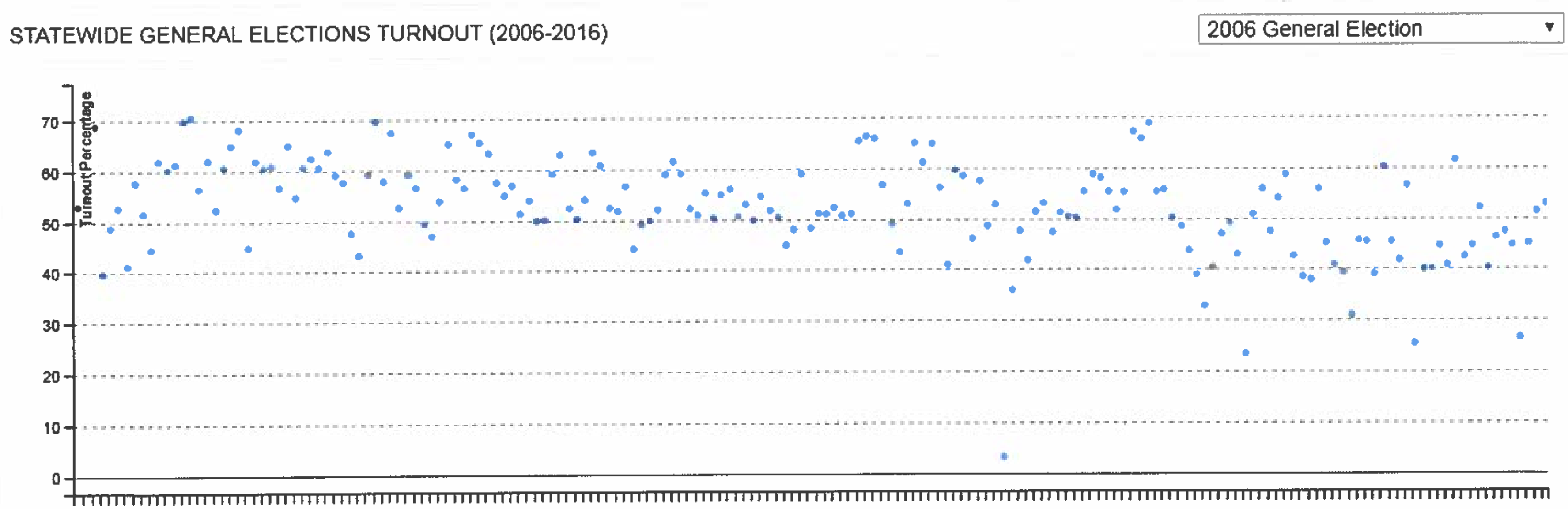
South Carolina 2016 General Election

CHARLESTON COUNTY

Participating Voters Demographics

Precinct	Precinct Code	Total Registered	Total Voting	Percent Voting	White	Nonwhite	Male	Female	Unknown Gender	18-24	25-44	45-64	65-up
AWENDAW	004	1299	926	71.285	478	448	417	509	0	33	194	421	278
CHRIST CHURCH	008	927	667	71.952	376	291	303	363	1	26	149	320	172
ISLE OF PALMS 1A	010	1088	776	71.323	757	19	386	390	0	30	164	323	259
ISLE OF PALMS 1B	011	1443	1007	69.785	988	19	484	523	0	44	235	485	243
ISLE OF PALMS 1C	012	1573	1220	77.558	1202	18	570	650	0	36	138	396	650
SULLIVANS ISLAND	032	1662	1236	74.368	1221	15	581	655	0	40	281	557	358
MCCLELLANVILLE	100	1677	1122	66.905	487	635	466	656	0	23	294	417	388
DEER PARK 1A	102	1380	709	51.376	303	406	261	448	0	35	353	240	81
DEER PARK 1B	103	3041	1613	53.041	1082	531	656	957	0	51	669	415	478
DEER PARK 2A	104	2948	1557	52.815	911	646	628	929	0	38	675	547	297
DEER PARK 2B	105	1925	998	51.844	518	480	418	580	0	30	525	339	104
DEER PARK 2C	106	1049	673	64.156	427	246	287	386	0	18	202	239	214
DEER PARK 3	107	1895	1100	58.047	487	613	464	636	0	39	494	405	162
LINCOLNVILLE	110	1513	961	63.516	577	384	387	574	0	21	284	346	310
LADSON	132	2850	1595	55.964	811	784	646	949	0	52	621	617	305
EDISTO ISLAND	173	1480	1127	76.148	715	412	494	633	0	24	199	443	461
WADMALAW ISLAND 1	174	1109	810	73.038	505	305	370	440	0	21	193	355	241
WADMALAW ISLAND 2	175	1170	860	73.504	371	489	376	484	0	27	183	324	326
KIAWAH ISLAND	180	1683	1280	76.054	1258	22	603	676	1	10	45	367	858
TOWN OF SEABROOK	181	1945	1478	75.989	1444	34	700	778	0	12	67	431	968
FOLLY BEACH 1	184	1084	693	63.929	677	16	341	352	0	15	200	297	181
FOLLY BEACH 2	185	1256	868	69.108	858	10	424	444	0	16	175	350	327
CHARLESTON 1	201	872	647	74.197	638	9	305	342	0	14	135	223	275
CHARLESTON 2	202	979	664	67.824	654	10	321	343	0	14	127	198	325
CHARLESTON 3	203	1118	770	68.872	753	17	353	417	0	21	228	281	240
CHARLESTON 4	204	1065	598	56.15	529	69	243	355	0	20	269	178	131
CHARLESTON 5	205	1030	594	57.669	580	14	258	336	0	15	264	163	152
CHARLESTON 6	206	1450	737	50.827	601	136	289	448	0	36	260	165	276
CHARLESTON 7	207	1690	789	46.686	724	65	345	444	0	114	301	159	215
CHARLESTON 8	208	1249	659	52.762	315	344	304	355	0	27	373	170	89

The New Way

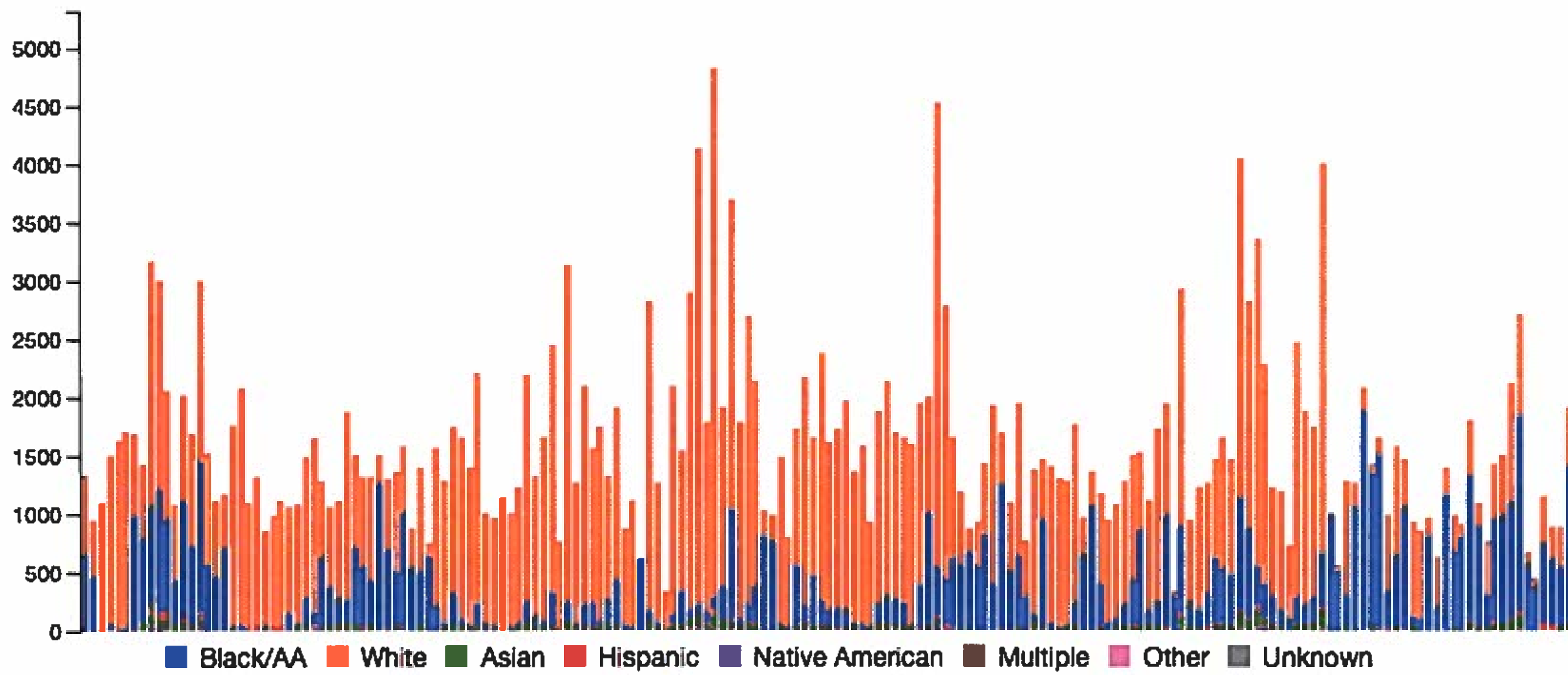

STATEWIDE GENERAL ELECTIONS TURNOUT (2006-2016)
2006 General Election
Turnout Percentage
70
60
50
40
30
20
10
0

The Old Way

	A	B	C	D	E	F	G	H	I	J	K
1	first_column	Total	Black/AA	White	Asian	Hispanic	Native American	Multiple	Other	Unknown	
2	Awendaw	1350	651	686	3	5	1	1	3	0	
3	Christ Church	945	438	495	3	2	1	1	4	1	
4	Isle Of Palms 1A	1108	6	1082	10	4	2	0	3	1	
5	Isle Of Palms 1B	1500	5	1462	10	4	4	3	10	2	
6	Isle Of Palms 1C	1644	6	1614	9	6	1	0	6	2	
7	Sullivans Island	1710	7	1685	5	10	0	0	2	1	
8	McClellanville	1688	961	707	4	2	1	3	9	1	
9	Deer Park 1A	1442	689	654	26	43	6	2	22	0	
10	Deer Park 1B	3177	862	2104	56	96	14	6	36	3	
11	Deer Park 2A	3017	1032	1802	61	74	8	7	29	4	
12	Deer Park 2B	2070	775	1115	63	72	10	8	27	0	
13	Deer Park 2C	1083	349	663	29	26	2	3	11	0	
14	Deer Park 3	2027	960	932	45	67	2	3	18	0	
15	Lincolnville	1696	643	984	15	28	2	1	21	2	
16	Ladson	3011	1275	1556	43	83	6	3	43	2	
17	Edisto Island	1519	553	951	3	3	5	0	2	2	
18	Wadmalaw Island 1	1128	434	675	2	9	1	1	6	0	
19	Wadmalaw Island 2	1175	680	483	2	2	1	2	4	1	
20	Kiawah Island	1767	6	1734	12	0	4	0	7	4	
21	Town Of Seabrook	2086	27	2037	6	5	0	0	9	2	
22	Folly Beach 1	1105	8	1073	5	7	3	2	7	0	
23	Folly Beach 2	1316	4	1288	6	6	1	1	7	3	
24	Charleston 1	860	5	838	5	6	0	1	5	0	
25	Charleston 2	996	1	975	2	8	0	0	9	1	
26	Charleston 3	1129	5	1103	6	8	1	0	5	1	
27	Charleston 4	1058	125	910	6	6	0	1	9	1	
28	Charleston 5	1079	9	1039	17	3	1	0	9	1	
29	Charleston 6	1483	217	1216	18	12	2	4	9	5	
30	Charleston 7	1653	100	1488	14	30	0	2	18	1	
31	Charleston 8	1289	616	646	8	9	0	2	7	1	
32	Charleston 9	1064	329	711	8	5	0	1	7	3	
33	Charleston 10	1115	204	844	23	15	1	6	19	3	
34	Charleston 11	1874	160	1628	33	22	5	2	21	3	
35	Charleston 12	1509	655	799	21	18	0	2	11	3	

The New Way

PRECINCT DEMOGRAPHICS



C3js Bar Chart

```
// Precinct Demographics Data
$(function () {
var chart4 = c3.generate({
    bindto: '#election-stat-chart-04',
           /*onrendered: function() {
                      d3.selectAll(".c3-axis.c3-axis-x .tick text")
                                 .style("display", "none");
           },*/
    data: {
        url: '/demographics.json',
        mimeType: 'json',
                      columns: 'x',
        keys: {
            x: 'first_column',
            value: [/*'Total',*/'Black/AA','White','Asian','Hispanic','Native American','Multiple','Other','Unknown']
        },
        type: 'bar',
            groups: [
            [/*'Total',*/'Black/AA','White','Asian','Hispanic','Native American','Multiple','Other','Unknown']
        ]
    },
    axis: {
       x: {
            type: 'category',
                                 categories: ['Awendaw','Charleston 1', …….
```

Demographics.json


```
[
	{
		"first_column": "Awendaw",
		"Total": "1350",
		"Black/AA": "651",
		"White": "686",
		"Asian": "3",
		"Hispanic": "5",
		"Native American": "1",
		"Multiple": "1",
		"Other": "3",
		"Unknown": "0"
	},
	{
		"first_column": "Christ Church",
		"Total": "945",
		"Black/AA": "438",
		"White": "495",
		"Asian": "3",
		"Hispanic": "2",
		"Native American": "1",
		"Multiple": "1",
		"Other": "4",
		"Unknown": "1"
	},
```

importJSON.php

```php
<?php
  header('Content-Type: application/json');
  $con=mysqli_connect("localhost", "votedata_SnrUsr1", "RD&AB~JmoI@b", "votedata_rsltData");

  // Check connection
  if (mysqli_connect_errno($con))
  {
    echo "Failed to connect to DataBase: " . mysqli_connect_error();
  }else
    {
      $data_points = array();
      $result = mysqli_query($con, "SELECT * FROM Precincts");
      while($row = mysqli_fetch_array($result))
    {
  $point = array("label" => $row['x'] , "y" => $row['y']);
    array_push($data_points, $point);
  }
  echo json_encode($data_points, JSON_NUMERIC_CHECK);
  }
  mysqli_close($con);
?>
```

Relationships

County: CHARLESTON
Filter by: Moved Into County
View Report
Start Date: 1/1/2017
End Date: 11/20/2017

1 of 130 Find | Next

RP0126

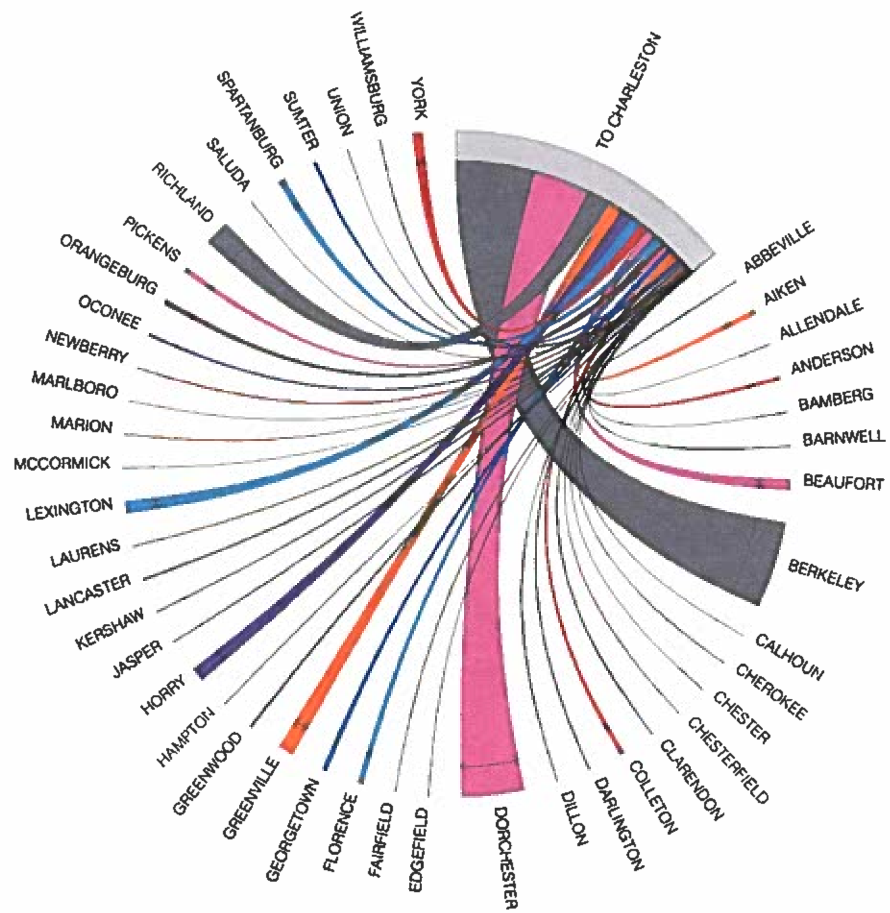
Voters Moved Into CHARLESTON County

Between 1/1/2017 and 11/20/2017

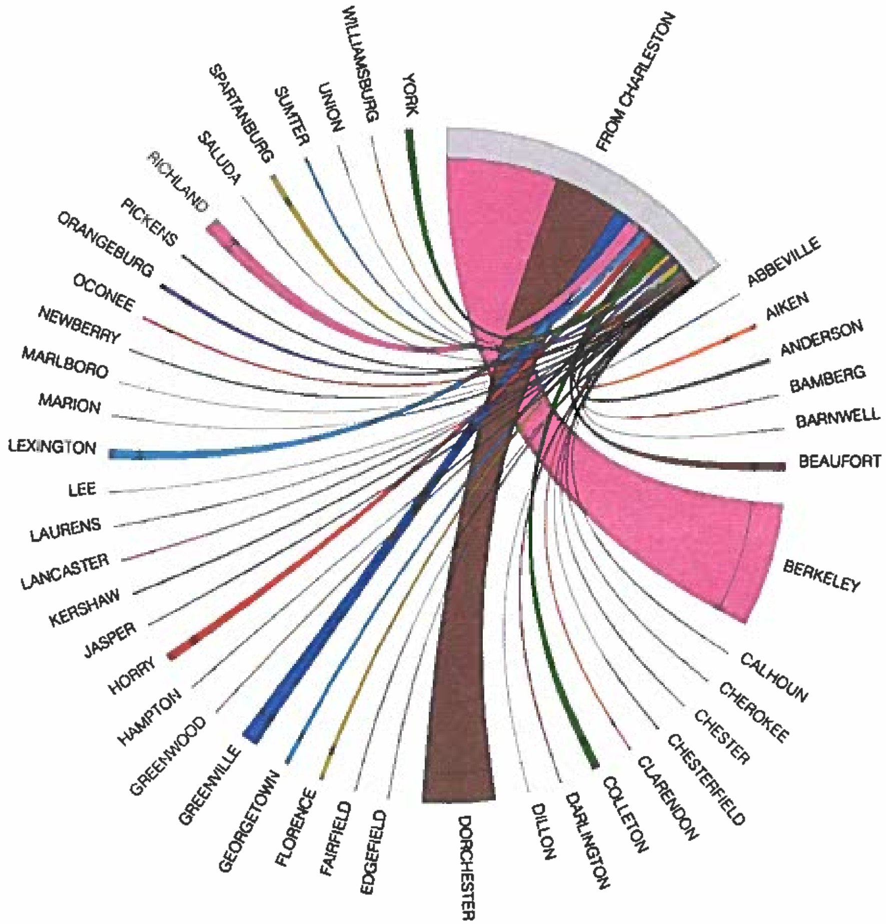
County: 10

Voter Name	Former County	Current Address	Former Address	Address End Date	VRN	Registration Date
Abbott, Alison M	18-DORCHESTER	322 Seneca River Dr Summerville SC 29485	109 Instructor Ct Ladson SC 29456	10/13/2017	084591478	10/12/2017
Abbott, Jonathan M	18-DORCHESTER	322 Seneca River Dr Summerville SC 29485	9018 Parlor Dr Ladson SC 29456	10/13/2017	185261831	10/12/2017
Abell, Jeremiah D	08-BERKELEY	983 Cassequo Province Mount Pleasant SC 29464	3005 Haswell St Apt 1315 Charleston SC 29492	7/3/2017	104882431	7/3/2017
Able, William J	18-DORCHESTER	4202 Montana Ave Apt B Charleston Afb SC 29404	117 Lancelot Hall North Charleston SC 29418	7/6/2017	470632601	6/27/2017
Abraham, Davarius A	40-RICHLAND	1947 Treebark Dr Charleston SC 29414	206 Sandpine Cir Columbia SC 29229	11/14/2017	406686615	11/6/2017
Abraham, Gerald C	40-RICHLAND	1002 Main St Charleston SC 29407	206 Sandpine Cir Columbia SC 29229	9/27/2017	406612690	9/19/2017
Ackerman, Michael S	08-BERKELEY	1533 Rainbow Rd Charleston SC 29412	365 Kelsey Blvd Wando SC 29492	4/7/2017	104911906	4/7/2017
Ackermann, Stephanie M	08-BERKELEY	4587 Lowell Dr Apt 3703 North Charleston SC 29418	537 Helmet Ave Cross SC 29436	7/7/2017	084556266	10/28/2017
Ackermann, Stephanie M	08-BERKELEY	4587 Lowell Dr Apt 3703 North Charleston SC 29418	538 Helmet Ave Cross SC 29436	11/8/2017	084556266	10/28/2017
Ackland, Lorri R	08-BERKELEY	8601 Dorchester Rd Apt 89 North Charleston SC 29418	121 Bull Run Dr Summerville SC 29486	4/19/2017	084604035	4/17/2017
Acorn, Lori O	21-FLORENCE	3040 Queensgate Way Mount Pleasant SC 29466	872 Chaucer Dr Florence SC 29505	5/31/2017	470002835	5/12/2017
Adair, Jennifer A	08-BERKELEY	1176 Manor Ln Mount Pleasant SC 29464	106 Sandshell Dr Charleston SC 29492	3/29/2017	406796699	3/17/2017
Adam, Callie E	39-PICKENS	107 Lochaven Dr Apt 204 Charleston SC 29414	291 Camp Creek Rd Central SC 29630	6/8/2017	396595551	6/8/2017
Adams Cowart, Teresa L	08-BERKELEY	9878 Tremont Ave Ladson SC 29456	6220 Murray Dr Apt 17G Hanahan SC 29410	5/10/2017	470592470	5/5/2017
Adams Samuels, Zachary Q	40-RICHLAND	63 Beaufain St Apt E Charleston SC 29401	3028 Pavilion Tower Cir Columbia SC 29201	8/24/2017	470280455	9/13/2017
Adams Samuels, Zachary Q	40-RICHLAND	63 Beaufain St Apt E Charleston SC 29401	3028 Vista Towers Dr Columbia SC 29201	9/20/2017	470280455	9/13/2017
Adams, Craig E	32-LEXINGTON	3773 Maidstone Dr Mount Pleasant SC 29466	242 Waterstone Dr Lexington SC 29072	6/29/2017	104940155	6/15/2017

Relationships



Relationships



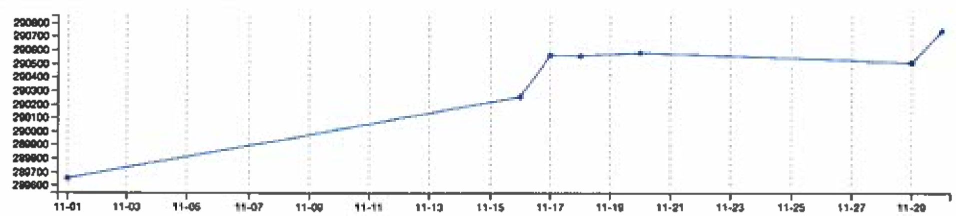
dataMining.js

```
var voterCSV = "votersToChas.csv";
var margin      = {top: 50, right: 50, bottom: 50, left: 50},
           width       = 1080 - margin.left - margin.right,
           height      = 768 - margin.top  - margin.bottom,
           innerRadius = Math.min(width, height) * .35,
           outerRadius = innerRadius * 1.1;

var svg = d3.select("#chordChart").append("svg")
        .attr("width",  width  + margin.left + margin.right)
        .attr("height", height + margin.top  + margin.bottom)
        .append("g")
        .attr("transform", "translate(" + margin.left + "," + margin.top + ")")
        .append("g")
        .attr("class", "chordgraph")
        .attr("transform", "translate(" + width/2 + "," + height/2 + ")");

    d3.csv(voterCSV, function(d){

           /*
            * IMPORTANT! You must specify the first column of data
            */
        var firstColumn = "first_column";
```

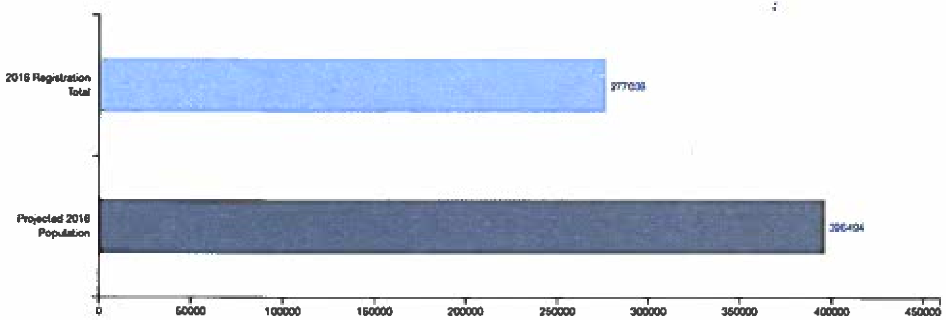
dataMining.js

```
//store column names
      var fc = d.map(function(d){ return d[firstColumn]; }),
          fo = fc.slice(0),
          maxtrix_size = (Object.keys(d[0]).length - 1) + fc.length,
          matrix  = [];

      //Create an empty square matrix of zero placeholders, the size of the ata
      for(var i=0; i < maxtrix_size; i++){
          matrix.push(new Array(maxtrix_size+1).join('0').split('').map(parseFloat));
      }

      //go through the data and convert all to numbers except "first_column"
      for(var i=0; i < d.length; i++){

          var j = d.length;//counter

          for(var prop in d[i]){
              if(prop != firstColumn){
                  fc.push(prop);
                  matrix[i][j] = +d[i][prop];
                  matrix[j][i] = +d[i][prop];
                  j++;
              }
          }
      }
```

DAILY VOTER REGISTRATION TOTALS



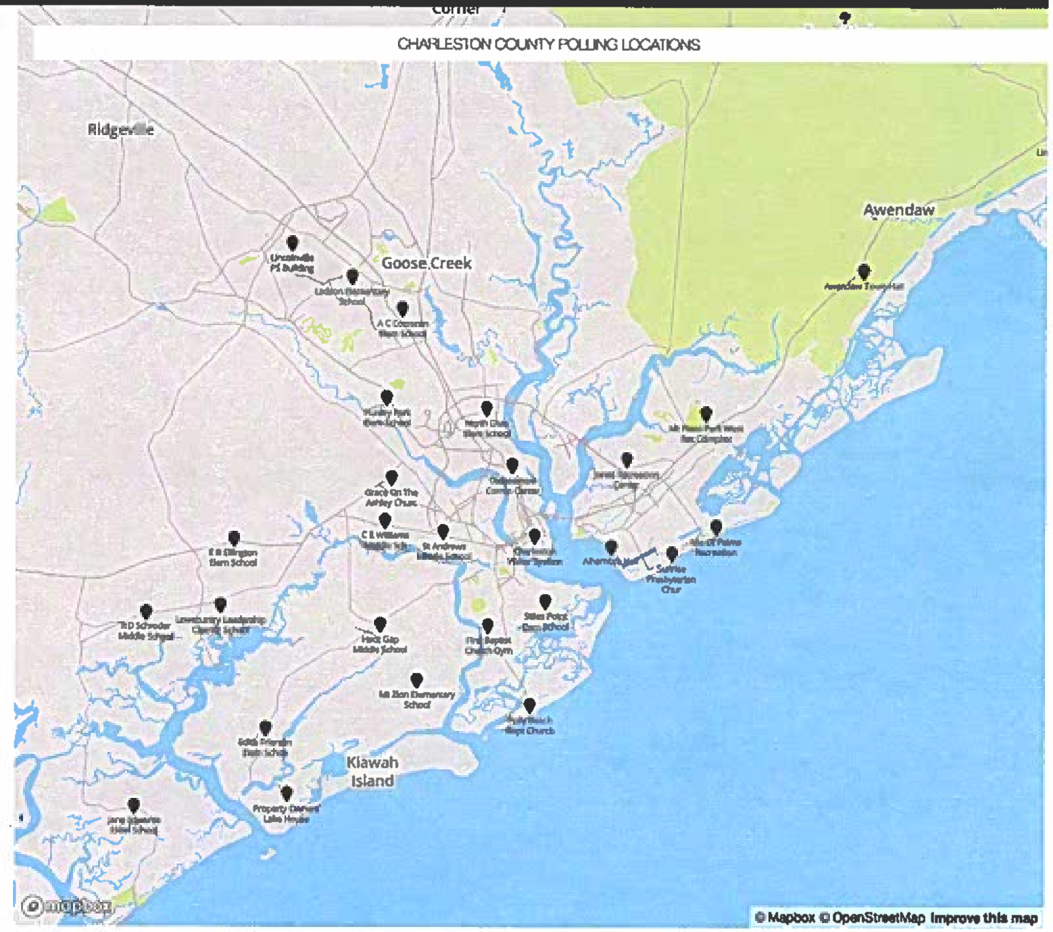
COMPARISON OF VOTER REGISTRATION VS. POPULATION (2016)



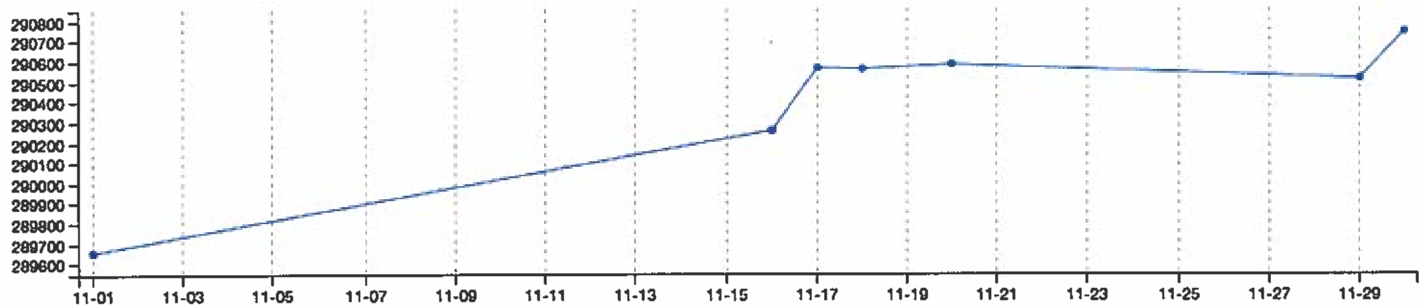
TIMELINE OF UPCOMING ELECTIONS

- Town of Awendaw Special Election

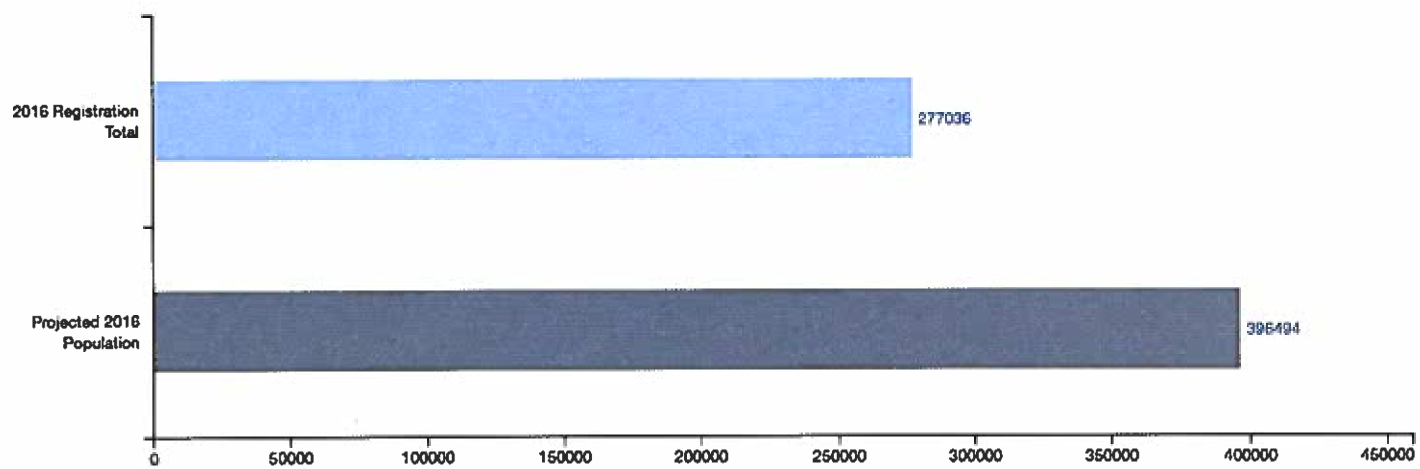
CHARLESTON COUNTY POLLING LOCATIONS



DAILY VOTER REGISTRATION TOTALS



COMPARISON OF VOTER REGISTRATION VS. POPULATION (2016)



Charleston County Pollin...

LOCATE COUNTY PRECI...

Search an address or locate on map

Find address or place

Charleston County Pollin...

LOCATE COUNTY PRECI...

Search an address or locate on map

2621 ELISSA DR

Information | Directions

0.53 miles · 3 minutes

ZOOM TO FULL ROUTE

1. Start at 2625 Elissa Dr

2. Go northwest on **Elissa Dr** toward **Glendale Dr**
0.17 mi 1 minute

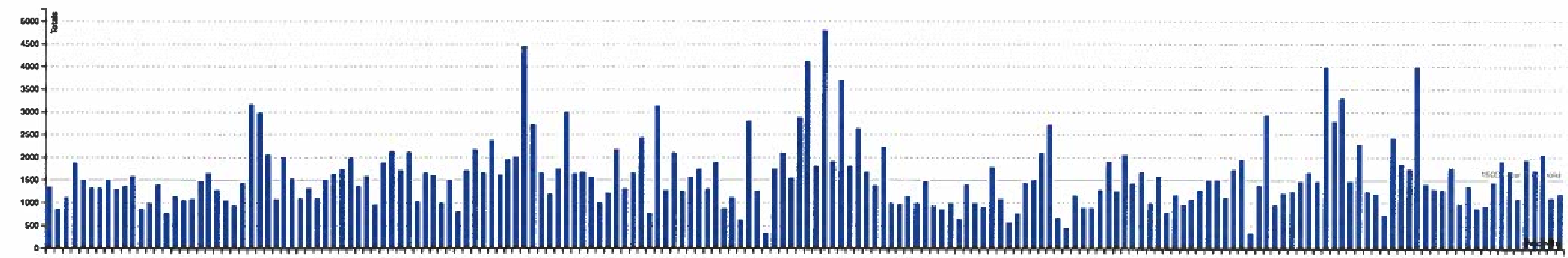
3. Turn right on **Glendale Dr**
0.13 mi 1 minute

4. Turn left on **Mona Ave**
0.1 mi

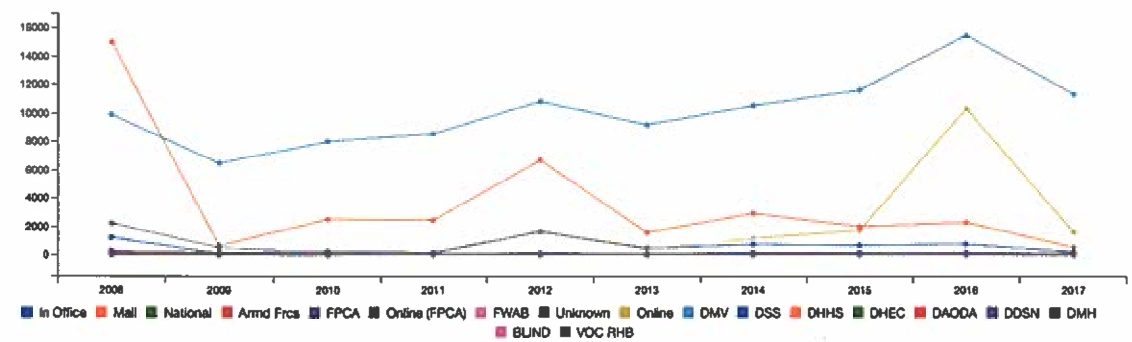
5. Turn left on **Wood Ave**
0.13 mi 1 minute

6. Finish at 2103 Wood Ave, on the left

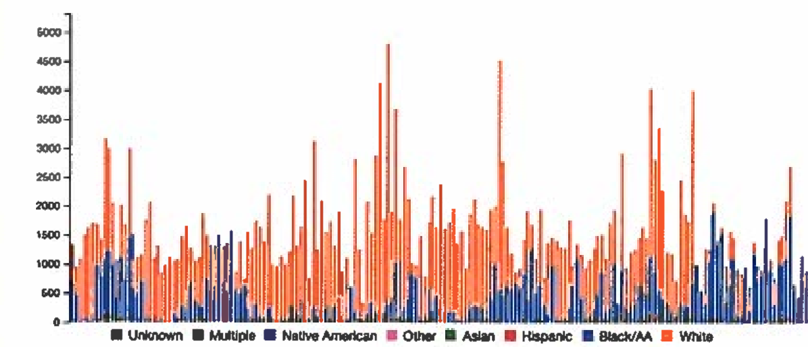
CURRENT REGISTRATION TOTALS BY PRECINCT



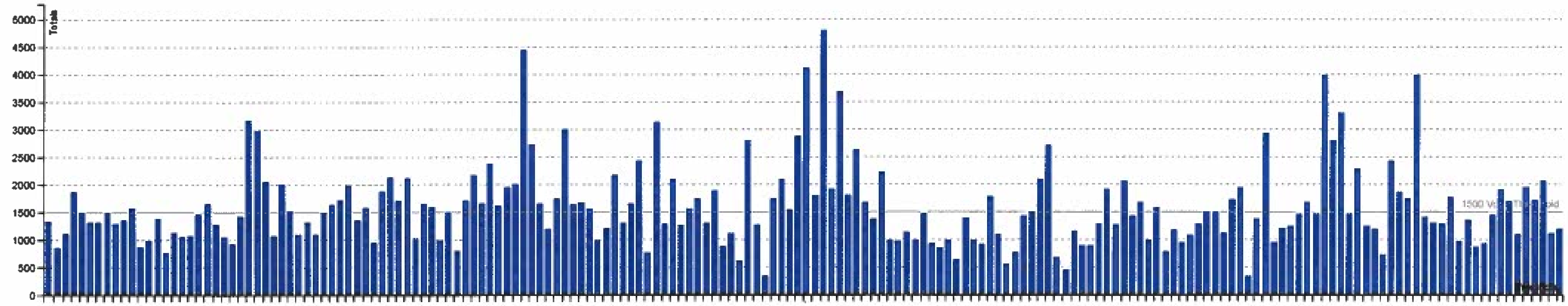
REGISTRATION LOCATION (HISTORICAL)



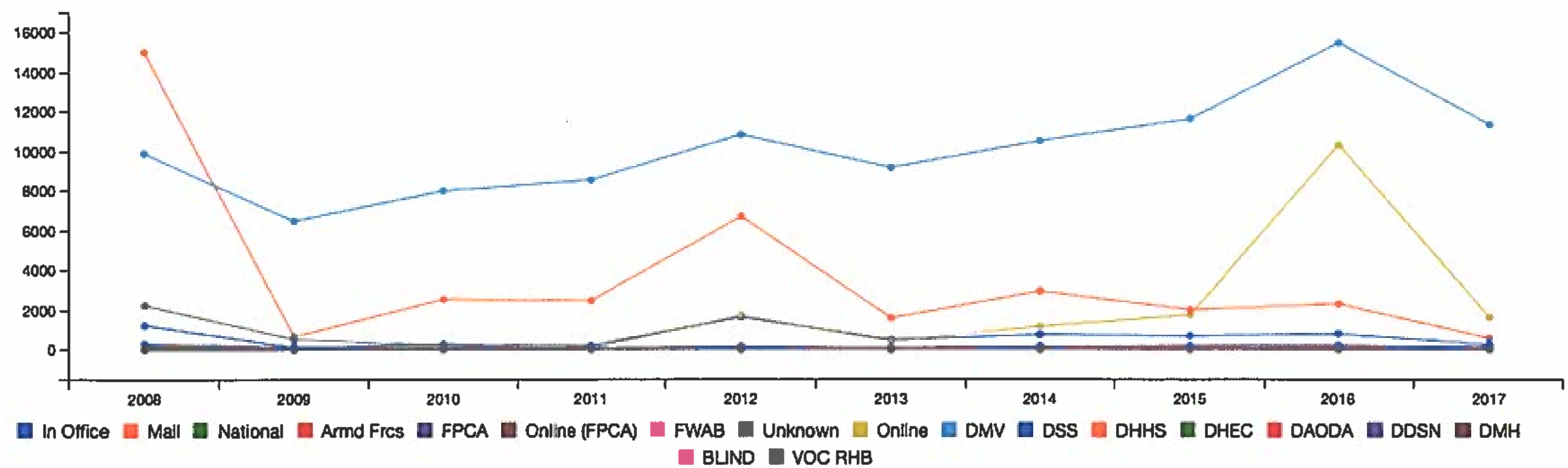
PRECINCT DEMOGRAPHICS



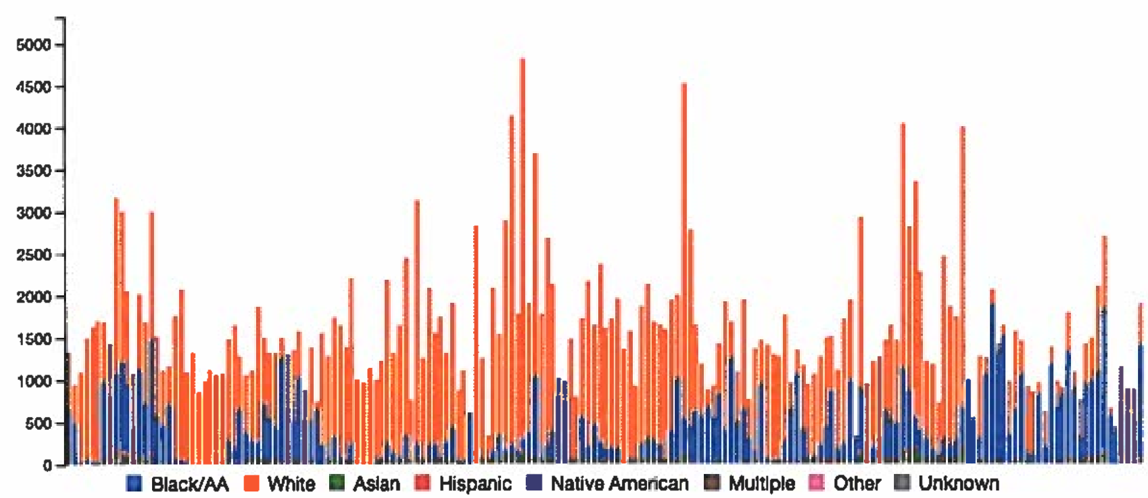
CURRENT REGISTRATION TOTALS BY PRECINCT



REGISTRATION LOCATION (HISTORICAL)

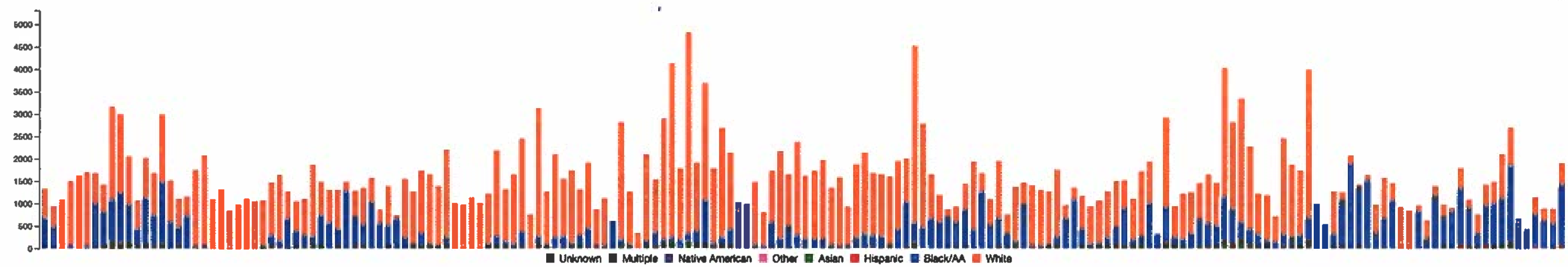


PRECINCT DEMOGRAPHICS



Dashboard Home Voter Statistics Voter Mapping Election Statistics Outreach Statistics Data Experiments

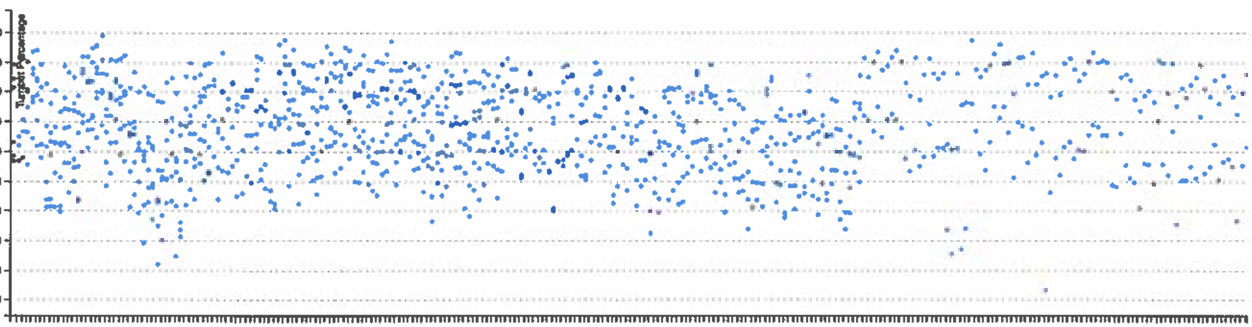
PRECINCT DEMOGRAPHICS



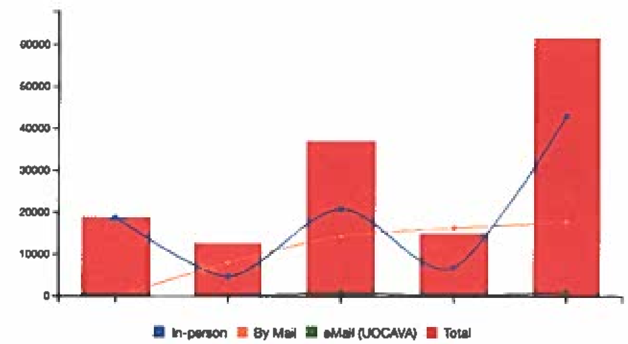
Dashboard Home | Voter Statistics | Voter Mapping | Election Statistics | Outreach Statistics | Data Experiments

STATEWIDE GENERAL ELECTIONS TURNOUT (2006-2016)

All General Elections (2006-2016)

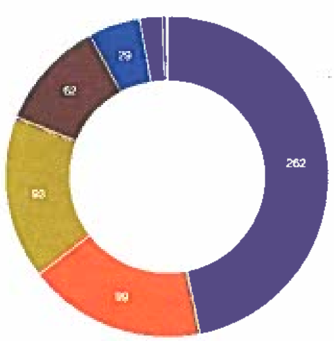


ABSENTEE VOTING METHOD COMPARISON

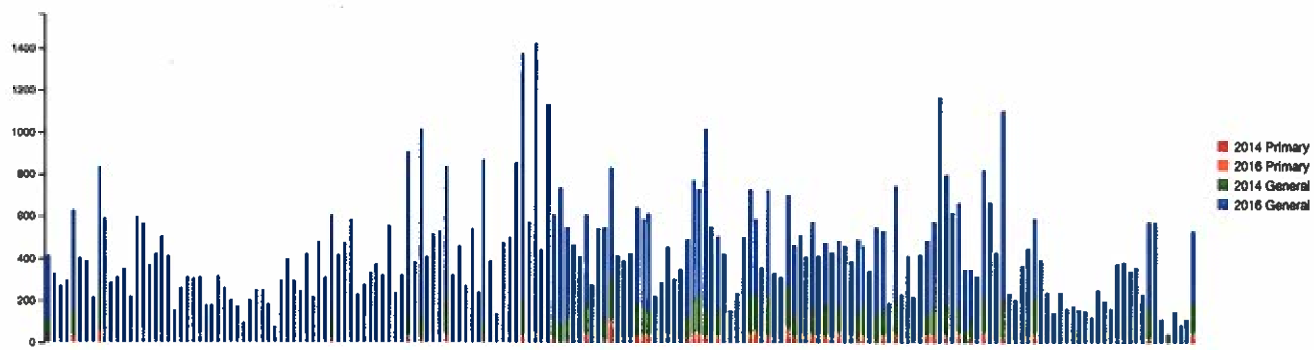


PROVISIONAL BALLOT STATISTICS (NOT COUNTED)

2016 General Election

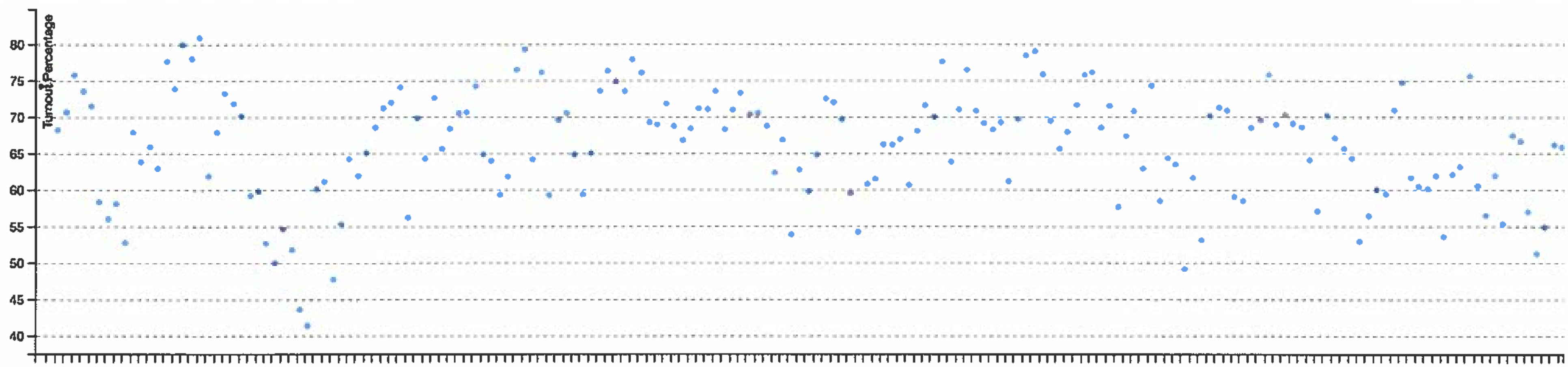


ABSENTEE BALLOTS BY PRECINCT



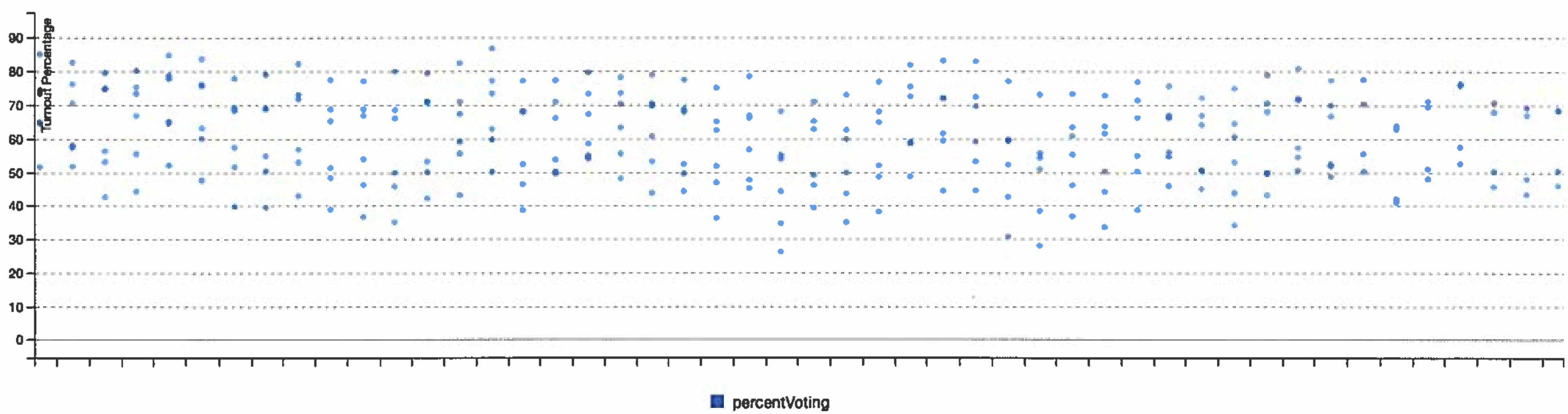
STATEWIDE GENERAL ELECTIONS TURNOUT (2006-2016)

2012 General Election

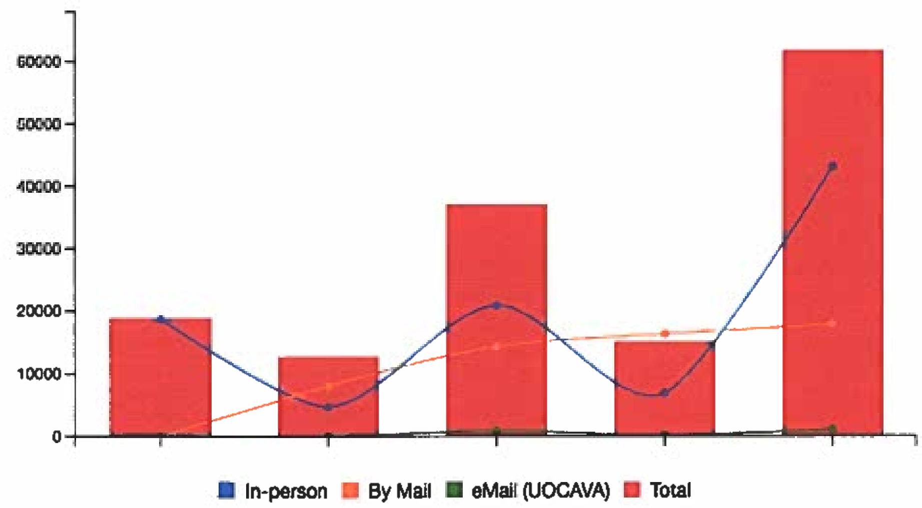


STATEWIDE GENERAL ELECTIONS TURNOUT (2006-2016)

All General Elections (2006-2016)



ABSENTEE VOTING METHOD COMPARISON



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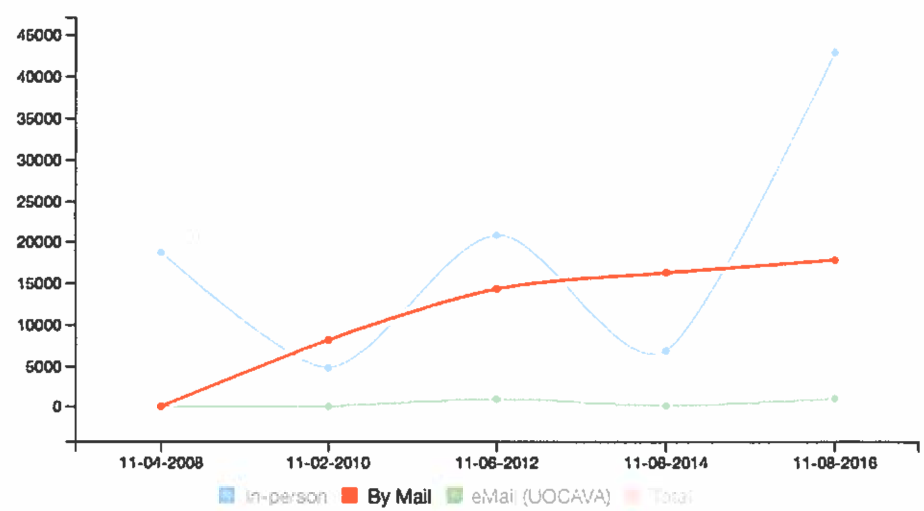
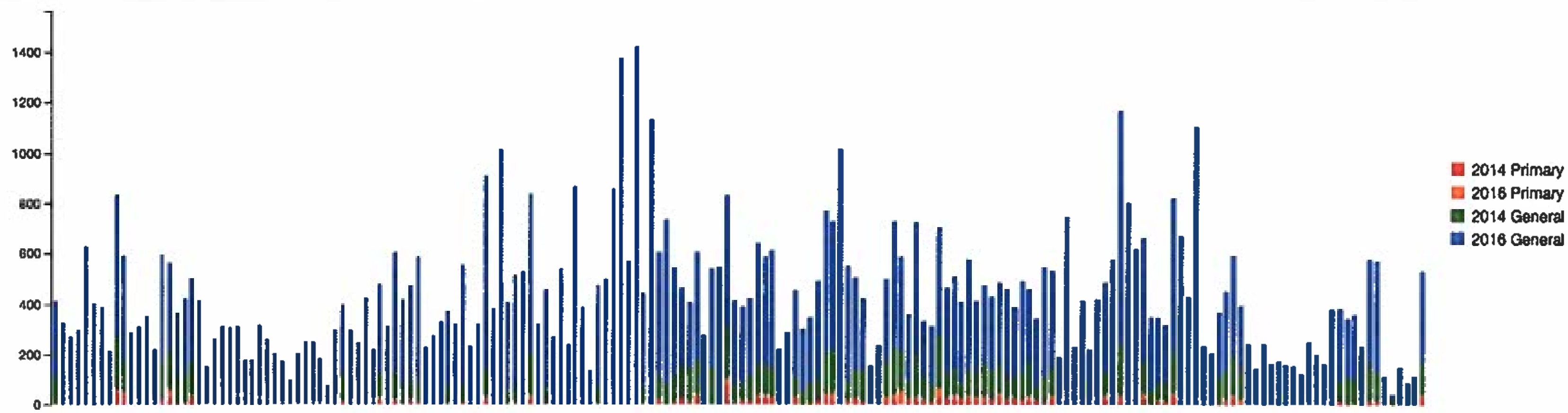
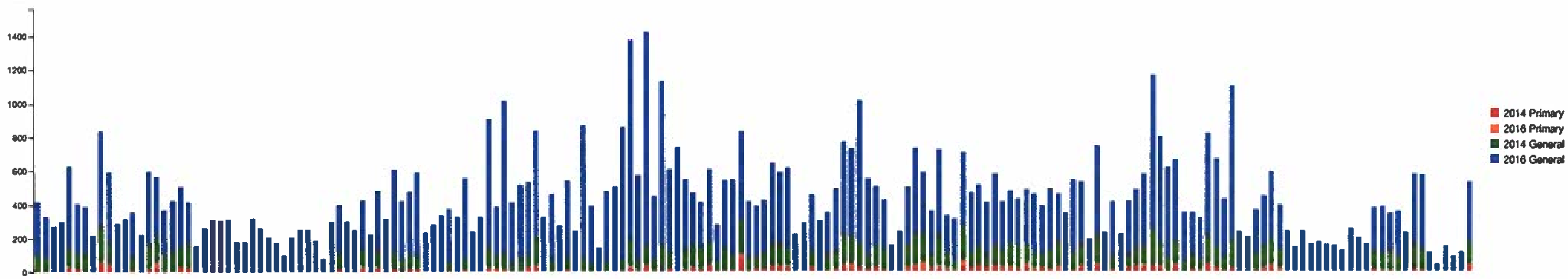


ABSENTEE BALLOTS BY PRECINCT



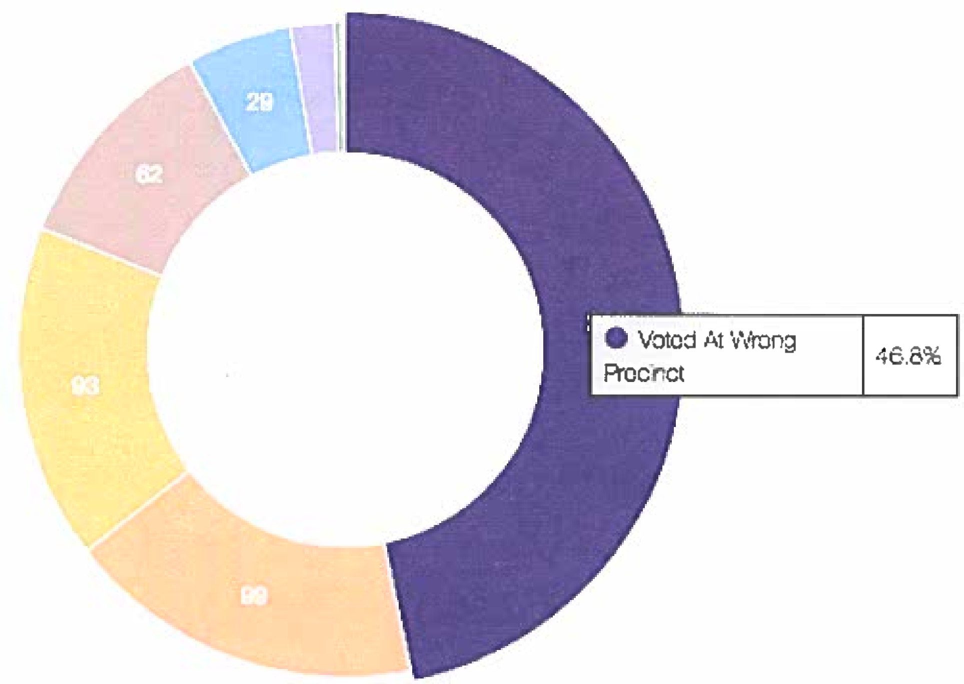
Dashboard Home Voter Statistics Voter Mapping Election Statistics Outreach Statistics Data Experiments

ABSENTEE BALLOTS BY PRECINCT



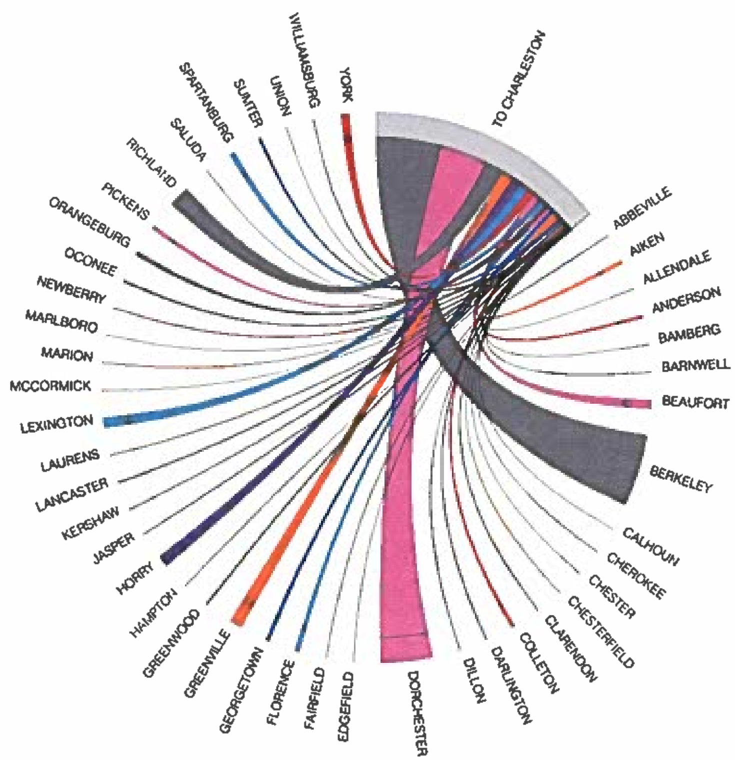
PROVISIONAL BALLOT STATISTICS (NOT COUNTED)

2016 General Election



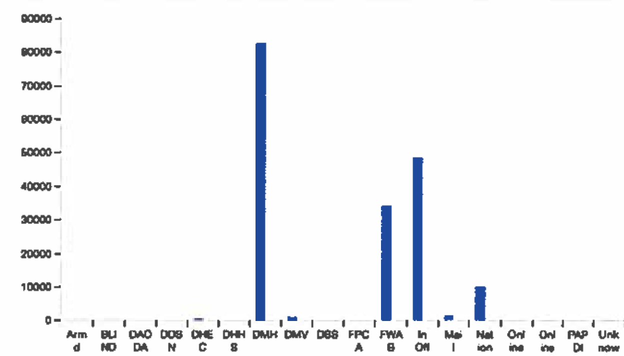
Dashboard Home | Voter Statistics | Voter Mapping | Election Statistics | Outreach Statistics | Data Experiments

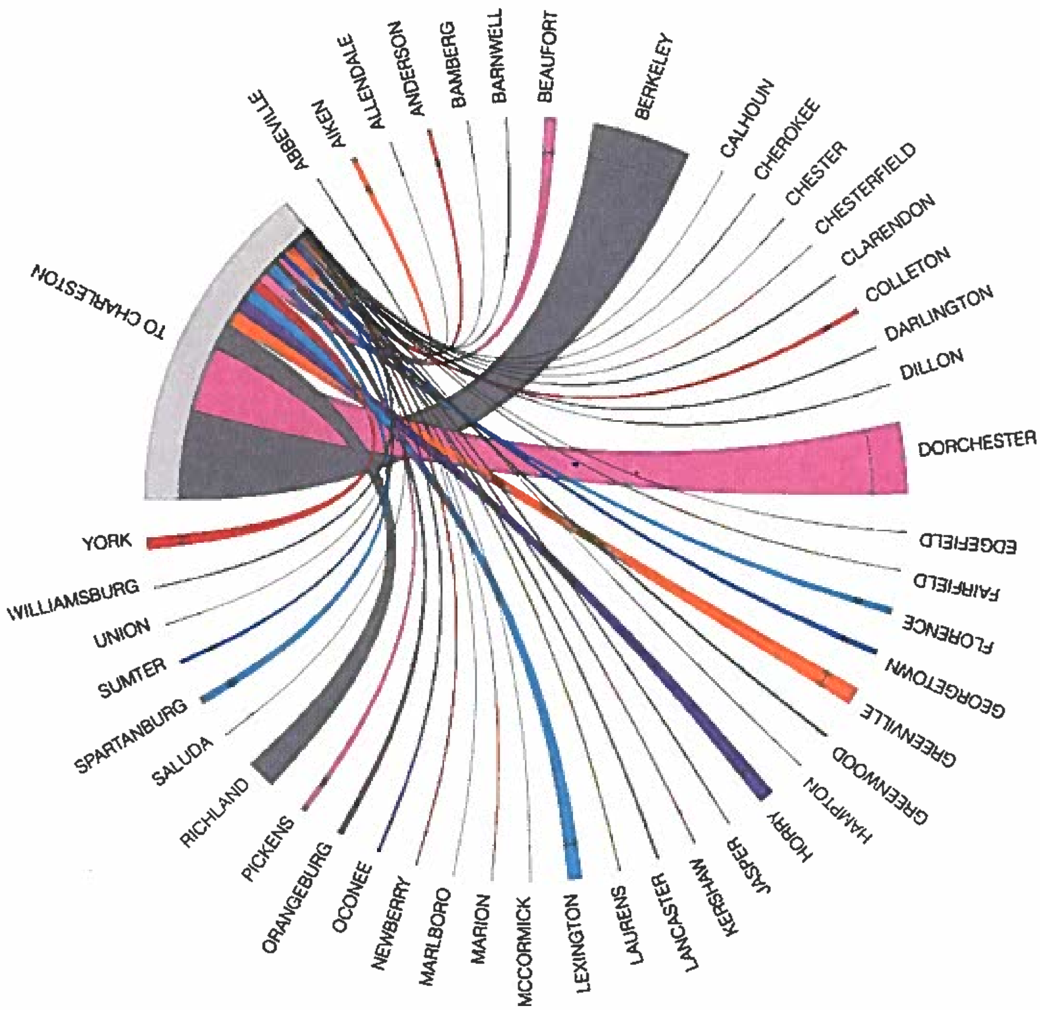
VOTERS MOVED TO CHARLESTON (2017)



PARTICIPATION BY REGISTRATION LOCATION

2016 General Election




TO CHARLESTON
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AIKEN
ALLENDALE
ANDERSON
BAMBERG
BARNWELL
BEAUFORT
BERKELEY
CALHOUN
CHEROKEE
CHESTER
CHESTERFIELD
CLARENDON
COLLETON
DARLINGTON
DILLON
DORCHESTER
EDGEFIELD
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GEORGETOWN
GREENVILLE
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MCCORMICK
MARION
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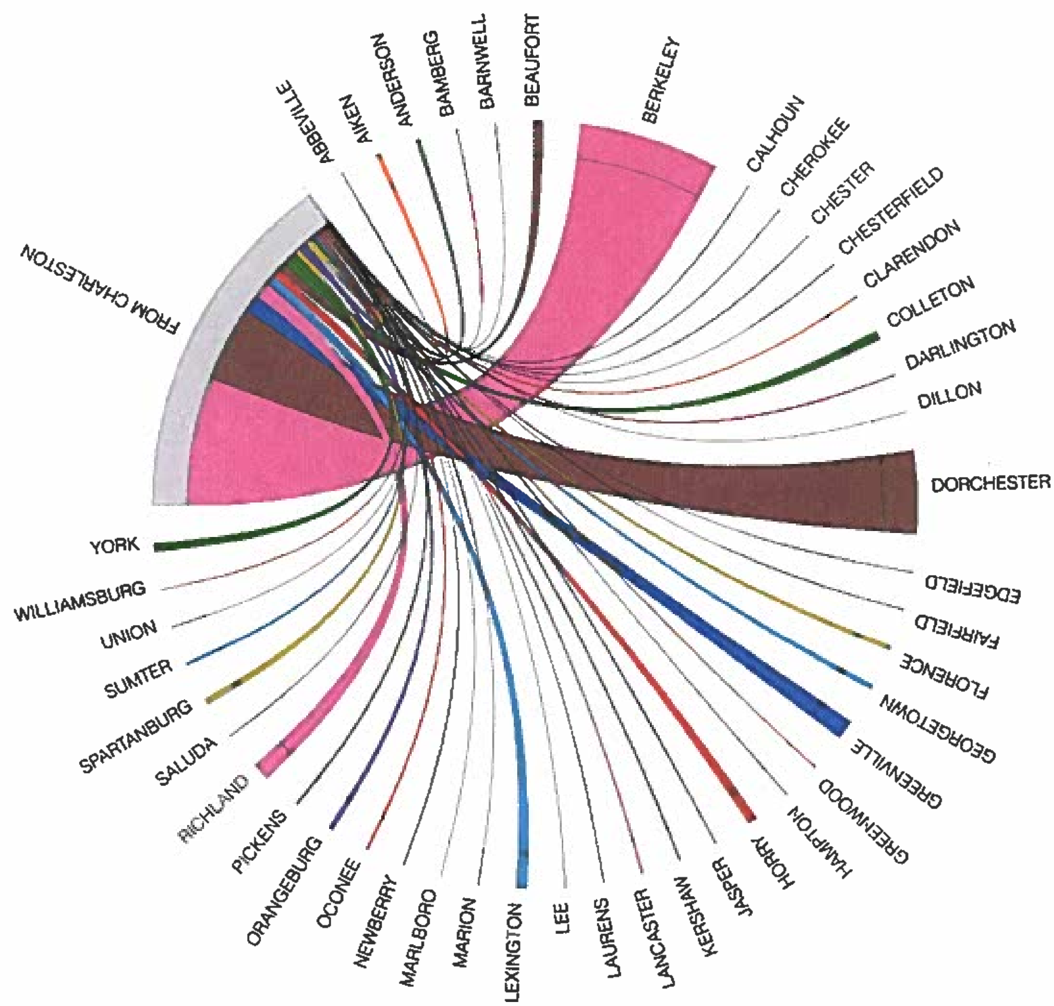
NEWBERRY
OCONEE
ORANGEBURG
PICKENS
RICHLAND
SALUDA
SPARTANBURG
SUMTER
UNION
WILLIAMSBURG
YORK



FROM CHARLESTON
ABBEVILLE
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BERKELEY
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CHEROKEE
CHESTER
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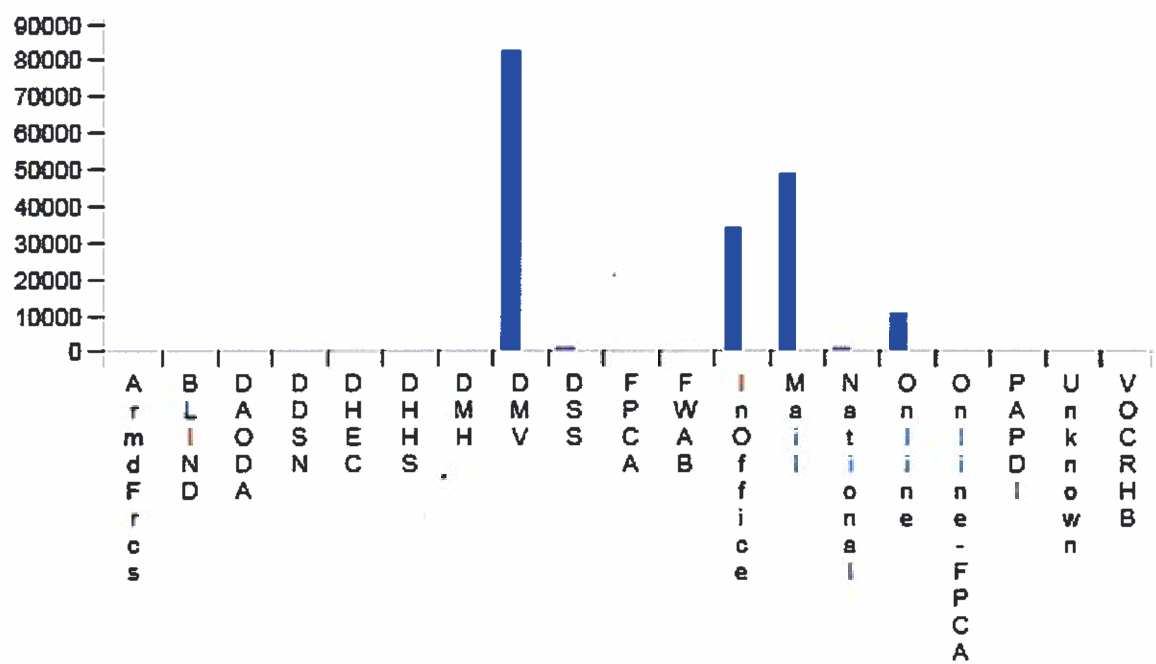
WILLIAMSBURG
YORK

PARTICIPATION BY REGISTRATION LOCATION

2016 General Election



Future Enhancements

- "Live" (or at least daily) data:
  - Absentee
  - Voter Registration Totals
- Expand data statewide
- Other Data mining techniques and experiments
- Include Voter Outreach statistics

Final Thoughts / Questions

Link

http://www.votedatacenter.us